

Article

Testing [O II] $\lambda 3727$ as a Star Formation Rate Tracer in Quasar Host Galaxies

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Abstract

The [O II] $\lambda 3727$ emission line is a widely used star formation rate (SFR) tracer. However, its application to type I quasars is not straightforward, because the line can be affected by dust extinction, metallicity and contamination from the AGN narrow-line region (NLR). We test the reliability of [O II] SFRs using a sample of 202 SDSS and PG quasars, by comparing [O II] SFRs and reference far-infrared (FIR) SFRs derived from multiwavelength SED decomposition. We measure [O II], [O III], and narrow Balmer emission lines by spectral fitting. Then, we calculate [O II] SFRs after correcting dust extinction and metallicity. We then compare these SFRs with the FIR SFRs, both with and without subtracting the AGN contribution estimated from [O III]. After this correction, the median offset between [O II] and FIR SFRs is -0.20 ± 0.72 dex for the full analysis sample and -0.17 ± 0.69 dex for sources with $S/N > 5$ in both [O II] and [O III]. Without subtracting the AGN contribution, the corresponding offsets are 0.00 ± 0.69 and 0.12 ± 0.66 dex. We conclude that [O II] is useful as a statistical SFR tracer for quasar host galaxies, but individual objects still require careful treatment of AGN contamination, extinction, metallicity, aperture effects, and redshift-dependent systematics.

Keywords: quasars; active galactic nuclei; quasar host galaxies; star formation rates; [O II] emission line; narrow-line region; optical spectroscopy

1. Introduction

Understanding the connection between supermassive black hole growth and star formation of the quasar host galaxies requires reliable measurements of the host-galaxy star formation rate (SFR). Far-infrared (FIR) emission is a reliable tracer of the star formation of quasar host galaxy, but deep FIR observations are not available for many large quasar samples. Optical spectra, in contrast, are available for large numbers of quasars from surveys such as the Sloan Digital Sky Survey (SDSS). If a robust optical emission-line SFR tracer can be applied to quasars, it would provide an efficient and complementary way to study star formation in quasar host galaxies.

FIR and optical emission-line indicators provide complementary measurements of star formation. When well-sampled FIR photometry and a reliable SED decomposition are available, FIR emission from the cold-dust component provides a useful reference



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SFR because it traces star formation that is obscured by dust. However, deep FIR observations are not available for most large quasar samples, and the relatively low angular resolution of FIR data can introduce blending and source-confusion uncertainties. In addition, separating the dust heated by star formation from the dust heated by AGN requires multiwavelength modeling.

In contrast, optical spectra are available for large quasar samples from surveys such as SDSS. The [O II] $\lambda 3727$ emission line is one of the most widely used optical SFR tracers for star-forming galaxies. The [O II] $\lambda 3727$ line is produced in H II regions ionized by young massive stars and can be measured directly from an optical spectrum over a broad redshift range [1,2]. Although the H α emission line is a more straight tracer, its wavelength is easy to be redshifted out of the observed optical wavelength range or not covered by the available spectra. [O II] $\lambda 3727$ has a shorter wavelength, making it accessible at higher redshift than H α in optical spectroscopic surveys. Therefore, [O II] is not expected to be more reliable than a carefully derived FIR SFR for individual quasars; its main advantage is that it can provide an observationally efficient SFR estimate for large spectroscopic samples when deep FIR data are unavailable. The purpose of this work is not to replace FIR SFRs, but to test whether [O II] can serve as a useful statistical SFR tracer for quasar host galaxies and to identify its main limitations. Unlike H α , [O II] is sensitive to dust attenuation, gas-phase metallicity, and ionization state. These dependencies introduce substantial scatter, so we need careful calibration to use [O II] as a quantitative SFR tracer [2,3].

But applying [O II] to AGNs and quasars is more challenging. In addition to star-forming H II regions, [O II] can also arise from AGN narrow-line region (NLR). Therefore, the total [O II] luminosity of a quasar is not a pure tracer of star formation. The [O II] SFR may be overestimated if we do not subtract the AGN contribution. This difficulty is especially important for type-I quasars. Because type-I quasars have a strong AGN continuum, broad emission lines, Fe emission, and narrow-line emission, they can influence the measurement of the emission features from the host galaxies.

Zhuang and Ho [4] recalibrated the relation between [O II] luminosity and SFR using a large sample of star-forming galaxies. They used a photoionization model to estimate how much the [O II] emission comes from the AGN NLR region. By subtracting the contribution from AGN NLR, the [O II] emission from host galaxies can be separated. After that, they found the corrected [O II] SFRs in good agreement with D4000 SFRs for a large sample of SDSS narrow-line AGNs. The median difference between these two SFRs is small, about 0.055 dex, though the scatter is large, about 0.335 dex. However, it is still unclear whether this method works well for luminous type-I quasars. In these quasars, the [O II] emission can be strongly affected by the luminous central AGN.

In this work, we investigate the reliability of [O II] $\lambda 3727$ as an SFR tracer for quasar host galaxies. We use a sample of 202 quasars, including SDSS quasars and PG quasars, with both optical spectra and FIR SFRs. The FIR SFRs are derived from multiwavelength SED decomposition and provide an independent estimation of SFR. We refit the SDSS spectra and newly obtained optical spectra for PG quasars to measure [O II] $\lambda 3727$, [O III] $\lambda 5007$ and narrow Balmer lines. We then derive [O II] SFRs after correcting by dust attenuation, metallicity and the AGN NLR contribution. By comparing the [O II] SFRs with FIR SFRs, we find that [O II] is useful as a statistical SFR tracer of quasar host galaxies, but it needs careful treatment of AGN contamination for individual objects. We also examine how the SFR offset depends on AGN activity, the infrared AGN fraction, and redshift, in order to identify possible sources of systematic uncertainty. The goal of this paper is not to investigate the FIR SFRs or dust properties themselves. Instead, the FIR SFRs are adopted as reference measurements for testing the [O II] emission line method. The derivation,

model dependence, and astrophysical implications of the FIR SFRs are discussed in a separate companion study.

This paper is organized as follows. Section 2 describes the quasar sample and the FIR reference SFRs. Section 3 presents the optical spectroscopic observations, spectral fitting, and emission-line measurements. Section 3.3 describes the derivation of [O II] SFRs. Section 4 compares the [O II] SFRs with FIR SFRs and discusses the reliability and limitations of the [O II] method. We summarize our main results in Section 5.

2. Sample

2.1. Sample Selection

Our quasar sample consists of two subsamples: one is selected from the SDSS DR7 quasar catalog [5–7] and the other is from the PG quasar sample [8,9]. Since the main goal of this work is to examine the reliability of the [O II] $\lambda 3727$ emission line as a SFR tracer of quasar host galaxies and compare it with independent FIR SFRs derived from multiwavelength SED decomposition, we require a quasar sample that has two parts of data: optical spectra covering the [O II] $\lambda 3727$ and [O III] $\lambda 5007$ emission lines, enough FIR photometric to constrain the FIR SED fitting to estimate FIR SFR.

For the SDSS DR7 quasar catalog, we restrict the redshift range to $0.02 < z < 0.8$ and then cross-match it with public Herschel photometric catalogs from the NASA/IPAC Infrared Science Archive (IRSA)¹. We require each source to be detected in at least two of the six Herschel bands (PACS 70, 100, and 160 μm ; SPIRE 250, 350, and 500 μm), so that the FIR SED can be meaningfully constrained. This selection results in 120 SDSS quasars with both optical spectra and FIR constraints.

PG quasars have been well studied and have plenty of spectroscopic and multiwavelength photometric data. In particular, the infrared photometry of 87 PG quasars has been compiled and modeled by Shangguan et al. [10]. These are optically bright, low-redshift ($z < 0.5$) quasars. Five of the 87 PG quasars overlap with the 120 SDSS quasars selected above. After removing these duplicates, the final sample contains 202 unique quasars.

2.2. Reference FIR Star Formation Rates

In this work, the FIR SFRs are used as reference values for [O II] SFRs to compare. A detailed analysis of the submillimeter observations, multiwavelength SED fitting, cold-dust templates, and host galaxy properties will be presented in a companion paper. Here, we only summarize the aspects that are required for the present comparison. The reference FIR SFRs were derived from multiwavelength SED decomposition of stellar, dust, AGN, and radio components. The FIR luminosity used for the SFR estimation is calculated from the cold-dust component integrated over 8–1000 μm in order to reduce contamination from AGN-heated dust. The adopted conversion is

$$\text{SFR}_{\text{FIR}} (M_{\odot} \text{ yr}^{-1}) = 3.02 \times 10^{-44} L_{\text{FIR}}^{\text{cold}} (\text{erg s}^{-1}), \quad (1)$$

where $L_{\text{FIR}}^{\text{cold}}$ is the infrared luminosity of the cold-dust integrated over 8–1000 μm . This conversion corresponds to the FIR SFR calibration of Kennicutt [1], rescaled to a Kroupa [11] initial mass function (IMF). Throughout this paper, all SFRs and stellar masses are applied to the same Kroupa IMF scale. The present paper does not attempt to interpret the FIR SFRs themselves, but uses them only to evaluate the reliability and limitations of [O II] SFR tracer.

3. Spectroscopic Observations and Measurements

3.1. [O II] $\lambda 3727$ Measurements for the SDSS Quasar Sample

For the 120 SDSS quasars in our sample, the catalog of Shen et al. [12] provides measurements of the continuum and a number of broad and narrow emission lines. These include the monochromatic luminosities at 1350, 3000, and 5100 Å, the broad and narrow components of H α and H β , and the narrow-line luminosities of [N II] $\lambda\lambda 6548, 6584$, [S II] $\lambda\lambda 6717, 6731$, and [O III] $\lambda\lambda 4959, 5007$. However, the [O II] $\lambda 3727$ line required for this work is not included in the catalog. We therefore refit the SDSS optical spectra to measure the [O II] $\lambda 3727$ luminosity.

The spectral fitting was performed with QSOFITMORE [13], a quasar spectral-fitting package designed to model both continuum and emission-line components. First, we correct the spectra for Galactic extinction using the dust map of Schlegel et al. [14]. Second, since the redshifts of the quasars are known, all the spectra are shifted to the rest frame. Third, we decompose each spectrum into host galaxy and quasar components using principal component analysis (PCA). Fourth, we fit the quasar continuum using Fe emission templates [8,15,16], a Balmer continuum component, and polynomial terms, using continuum windows that are relatively free from strong emission-line contamination. After subtracting the best-fit continuum, we fit the emission-line components. The H β and H α regions are fitted simultaneously with their neighboring narrow lines, including [O III] $\lambda\lambda 4959, 5007$, [N II] $\lambda\lambda 6548, 6584$, and [S II] $\lambda\lambda 6717, 6731$. Both H β and H α are modeled with broad and narrow components, while the forbidden lines are modeled as narrow components. For [O III] $\lambda\lambda 4959, 5007$, an additional blue-wing component is included when required. Each component is modeled with Gaussian profiles, the FWHM of the narrow components is constrained to be less than 800 km s^{−1}. The uncertainty of each fitted quantity is estimated from 50 Monte Carlo realizations.

Figure 1 shows an example of the spectral fitting for one SDSS quasar in our sample, J122102.50+155447.04. In the figure, the red curves represent the broad emission-line components, while the green curves represent the narrow emission-line components. For [O III] $\lambda\lambda 4959, 5007$, the green wide components on the blue side of the line profiles indicate the [O III] outflow-wing components. Although the main analysis adopts a single-Gaussian approximation for the unresolved [O II] doublet and one broad H β component, we assess the impact of these choices using an alternative fitting scheme for a high-quality validation subsample in Section 3.4. Residuals normalized by the spectral uncertainty are also shown to illustrate the fit quality.

The narrow emission line luminosity results of the SDSS quasars are listed in Appendix A Table A1. The table includes [O II], [O III], H α , and H β measurements for 115 SDSS quasars. The remaining five SDSS quasars overlap with the PG quasar sample and are listed together with the PG quasars in Section 3.2. Because the SDSS DR7 spectra cover only 3800–9200 Å, the H α emission line is shifted out for sources at $z \gtrsim 0.4$. So the H α luminosities are not available for these objects. For several quasars, the H α or H β emission is dominated by the broad component, so the corresponding narrow component is difficult to fit. In addition, some spectra have low signal-to-noise ratios or have low spectral quality in specific wavelength regions, so we do not get reliable measurements of certain emission lines. These unreliable measurements will be excluded from the subsequent analysis according to their signal-to-noise ratios.

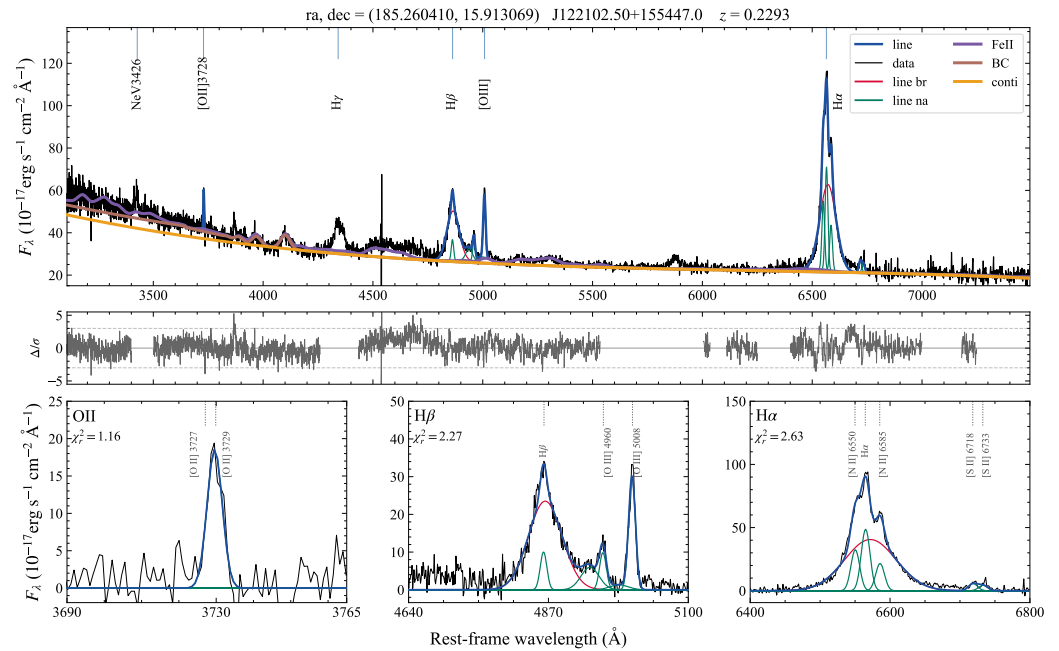


Figure 1. Example of the spectral fitting for the SDSS quasar J122102.50+155447.04 using QSOFITMORE. The black curve shows the observed spectrum. The blue curve shows the total fitted emission-line model. The purple, brown and orange curves represent the Fe II emission, the Balmer continuum, and the underlying continuum, respectively. The red and green curves denote the broad and narrow emission-line components, respectively. The unresolved [O II] feature is modeled with a single Gaussian profile and the broad H β profile is represented by one broad Gaussian component. For [O III] $\lambda\lambda$ 4959, 5007, the green wide components on the blue side of the line profiles indicate the [O III] outflow-wing components. The middle panel shows the residuals normalized by the spectral uncertainty $((F_{\text{data}} - F_{\text{model}})/\sigma)$ of the continuum and emission lines we fitted. It does not show the residuals of the emission lines we did not fit.

3.2. [O II] λ 3727 Measurements for the PG Quasar Sample and Observations

Among the 87 PG quasars, 31 have available SDSS spectra. We fit these spectra with QSOFITMORE using the same procedure described in Section 3.1. The resulting narrow-line luminosities of [O II], [O III], H α , and H β are listed in Appendix A Table A2. Two of them need additional notes. For PG1259+593, H α is not covered by the SDSS spectrum because of its high redshift. In addition, its H β region is dominated by the broad component, so the narrow H β and [O III] λ 5007 do not have reliable measurements. It also has no clear [O II] λ 3727 emission detected. For PG1415+451, the H β region is similarly dominated by the broad component, so the narrow H β component is not reliably measured.

For the remaining 56 PG quasars, we first collected publicly available optical spectra from the literature and archival databases. However, many of these spectra are not suitable for our analysis. Some were obtained several decades ago, some have low signal-to-noise ratios or lack associated noise spectra, and some have insufficient wavelength coverage. In particular, after correction to the rest frame, the blue end of many spectra does not cover the [O II] λ 3727 emission line. We therefore obtained new optical spectroscopic observations for these 56 PG quasars using several optical telescopes, including the Xinglong 2.16 m telescope (XLT), the Lijiang 2.4 m telescope (LJT), the Himalayan Chandra Telescope (HCT), and the Palomar P200/DBSP spectrograph.

In total, we obtain usable optical spectra for 52 sources. Although the spectra were initially flux-calibrated using spectrophotometric standard stars during the data reduction, we also need additional flux calibration because the observations were obtained with different instruments, under different weather conditions, and at different epochs. Consid-

ering the optical photometry used in the SED fitting is based mainly on SDSS photometry, to maintain consistency, we do the flux calibration using SDSS photometric magnitudes. For a few PG quasars without available SDSS photometry, we instead use Pan-STARRS1 photometry [17,18]. The flux-calibrated spectra are then fitted with QSOFITMORE. The key narrow emission lines luminosities and the instrument used for each source are listed in Appendix A Table A3. Missing entries are mainly caused by limited wavelength coverage, low spectral resolution, broad-line contamination or low signal-to-noise ratios at the blue end of the spectra, where [O II] $\lambda 3727$ is located.

3.3. [O II] Star Formation Rates

Zhuang and Ho [4] recalibrated the [O II] $\lambda 3727$ emission line SFR tracer by explicitly including the dependence on gas phase metallicity. We define the metallicity-dependent term as

$$C(x) = -4373.14 + 1463.92x - 163.045x^2 + 6.04285x^3, \quad (2)$$

where $x = 12 + \log(\text{O}/\text{H})$ is the gas-phase oxygen abundance. For star-forming galaxies, the [O II] SFR is then given by

$$\text{SFR}_{[\text{O II}]} = 5.3 \times 10^{-42} L_{[\text{O II}]} C(x)^{-1}. \quad (3)$$

For AGNs, Zhuang and Ho [4] proposed subtracting the narrow-line region contribution to [O II] using [O III] $\lambda 5007$. The AGN-corrected [O II] SFR is

$$\text{SFR}_{[\text{O II}]} = 5.3 \times 10^{-42} \left(L_{[\text{O II}]} - 0.109 L_{[\text{O III}]} \right) C(x)^{-1}. \quad (4)$$

Here, $\text{SFR}_{[\text{O II}]}$ is in units of $M_{\odot} \text{ yr}^{-1}$, and $L_{[\text{O II}]}$ and $L_{[\text{O III}]}$ are the extinction-corrected line luminosities in units of erg s^{-1} . The term $0.109 L_{[\text{O III}]}$ represents the expected [O II] contribution from the AGN narrow-line region, assuming that [O III] is dominated by AGN photoionization. The remaining [O II] luminosity is then attributed to star formation in the host galaxy. The [O II] SFR calibration adopted from Zhuang and Ho [4] is used on the same Kroupa IMF scale as the FIR reference SFRs. This ensures that the comparison between $\text{SFR}_{[\text{O II}]}$ and SFR_{FIR} is not affected by an IMF-dependent normalization offset.

For quasars with reliable measurements of both narrow $\text{H}\alpha$ and narrow $\text{H}\beta$, we correct the emission-line luminosities by using the Balmer decrement, adopting an intrinsic line ratio of $(\text{H}\alpha/\text{H}\beta)_{\text{int}} = 3.1$ and the Cardelli et al. [19] extinction curve with $R_V = 3.1$. For quasars without reliable narrow $\text{H}\alpha$ or $\text{H}\beta$ measurements or those with a Balmer decrement smaller than 3.1, we instead use the attenuation from the SED fitting to correct the line luminosities. We note that this SED extinction correction is used only when the Balmer decrement is unavailable, and it may introduce additional uncertainty because the SED attenuation is less directly tied to the narrow line. The gas-phase metallicity is estimated from the stellar mass–metallicity relation ($M_{\star}$ – Z relation) [20,21], using the stellar masses derived from the SED fitting.

We compute the [O II] SFRs using Equation (4). For comparison, we also calculate the SFRs using Equation (3) without subtracting the AGN NLR contribution. The object-by-object extinction corrections, corrected line luminosities, FIR reference SFRs, AGN-corrected [O II] SFRs, and [O II] SFRs without subtracting AGN contribution are provided in Appendix A Table A4. A small number of quasars among the 202 objects is not listed because either the [O II] or [O III] emission-line luminosity cannot be reliably measured. Therefore, the object numbers in Appendix A Table A4 are not consecutive. In the following analysis, we focus on the statistical comparison between the [O II] SFRs and the FIR reference SFRs.

3.4. Robustness of the Emission-Line Measurements

The main analysis models the unresolved [O II] feature with a single Gaussian and adopts a simplified decomposition for the broad H β or H α profile. To assess the influence of these assumptions, we selected 55 SDSS spectra with continuum S/N > 15 and refitted them with an alternative model. In the alternative fitting, [O II] was represented by a constrained $\lambda 3726, \lambda 3729$ doublet with a fixed wavelength separation and a common velocity width. The broad H β and H α profiles were modeled with two Gaussian components. The continuum treatment, Fe II template, narrow-line constraints, and treatment of the [O III] wing were otherwise kept unchanged.

Table 1 compares the results obtained with the two fitting procedures. The integrated [O II] luminosity is nearly unchanged, with a median difference of $-0.000^{+0.001}_{-0.005}$ dex. Therefore, for this high-quality validation subsample, the single-Gaussian approximation does not introduce a systematic bias in the total [O II] flux. The alternative broad-H β decomposition yields median changes of $-0.12^{+0.08}_{-0.16}$ dex in the narrow H β luminosity and $-0.05^{+0.05}_{-0.16}$ dex in the [O III] luminosity.

Table 1. Effect of the alternative emission-line fitting model on the derived quantities for the high-quality validation subsample. The alternative model uses a constrained [O II] $\lambda 3726, \lambda 3729$ doublet and two broad H β and H α Gaussian components. The quoted values are the median differences, $\Delta Q = Q_{\text{alt}} - Q_{\text{orig}}$, with the 16th–84th percentile range.

Quantity	N	Median ΔQ	Unit
$\log L_{[\text{O II}]}$	55	$-0.000^{+0.001}_{-0.005}$	dex
$\log L_{[\text{O III}]}$	55	$-0.05^{+0.05}_{-0.16}$	dex
$\log L_{\text{H}\beta, \text{narrow}}$	55	$-0.12^{+0.08}_{-0.16}$	dex
$\log L_{\text{H}\alpha, \text{narrow}}$	29	$-0.17^{+0.11}_{-0.15}$	dex
$E(B - V)_{\text{Balmer}}$	29	$-0.05^{+0.24}_{-0.25}$	mag
$\log \text{SFR}_{[\text{O II}] - 0.109[\text{O III}]}$	55	$-0.08^{+0.38}_{-0.40}$	dex
$\log \lambda_{\text{Edd}}$	55	$0.11^{+0.10}_{-0.09}$	dex

Notes. The $E(B - V)_{\text{Balmer}}$ row includes only the sources for which the extinction correction is derived from the narrow-line Balmer decrement. For the remaining sources, the same SED-based attenuation prescription as in the main analysis is retained. The black-hole masses and Eddington ratios are recalculated using the FWHM of the composite broad H β profile, rather than from either individual Gaussian component.

For the 29 sources whose extinction correction is based on the narrow-line Balmer decrement, the median change in $E(B - V)$ is $-0.05^{+0.24}_{-0.25}$ mag. After propagation through the extinction correction and the $0.109L_{[\text{O III}]}$ subtraction, the AGN-corrected [O II] SFR changes by a median of $-0.08^{+0.38}_{-0.40}$ dex. Thus, the alternative decomposition can affect the derived SFR of individual quasars, but does not introduce a large systematic offset for the validation subsample. The alternative broad-H β model changes the single-epoch black-hole masses by a median of $-0.11^{+0.09}_{-0.10}$ dex and correspondingly increases the Eddington ratios by $0.11^{+0.10}_{-0.09}$ dex.

4. Comparison Between [O II] $\lambda 3727$ and FIR Star Formation Rates

4.1. Comparison Results

We compare the [O II] $\lambda 3727$ SFRs with the FIR SFRs derived from SED fitting in Figure 2. The abscissa in both panels is the FIR SFR. In the left panel, the ordinate in the upper subpanel is the [O II] SFR computed using Equation (4), where the AGN narrow-line contribution to [O II] is estimated as $0.109L_{[\text{O III}]}$ and subtracted from the total [O II] luminosity. In the right panel, the ordinate in the upper subpanel is the [O II] SFR computed directly from the total [O II] luminosity using Equation (3), without subtracting the $0.109L_{[\text{O III}]}$ term. In both panels, the lower subpanel shows the difference between the

two SFRs, $\log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}}$. Blue points denote sources with $S/N > 5$ in both the $[\text{O II}]$ and $[\text{O III}]$ emission lines.

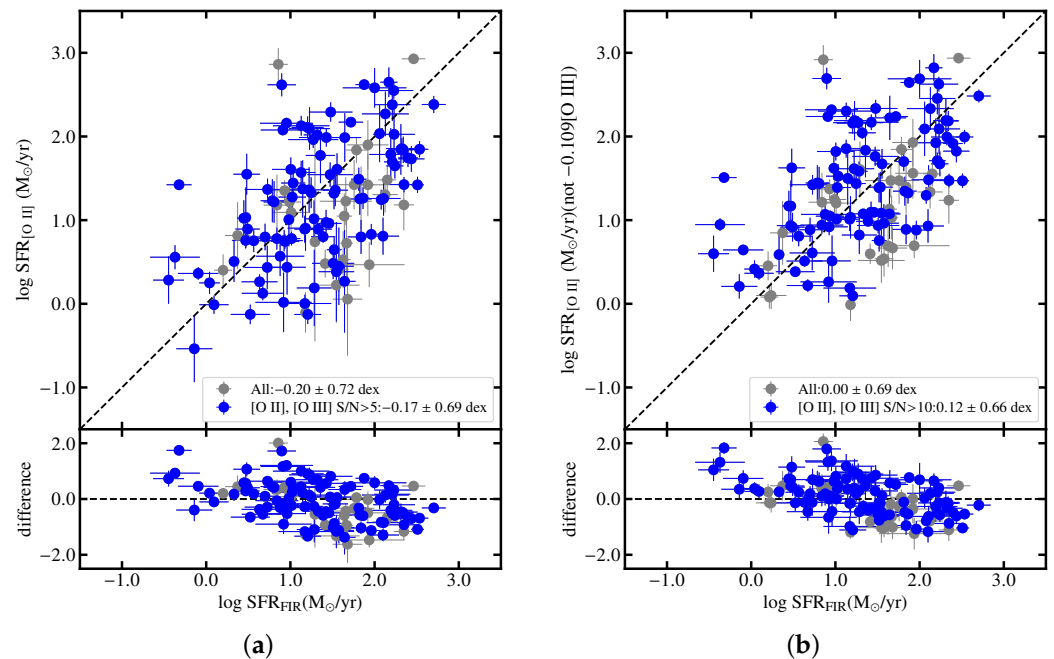


Figure 2. Comparison between $[\text{O II}] \lambda 3727$ SFRs and FIR SFRs. Panel (a) shows the $[\text{O II}]$ SFR after subtracting the AGN narrow-line contribution estimated as $0.109L_{[\text{O III}]}$, while panel (b) shows the $[\text{O II}]$ SFR computed without this correction. In both panels, the lower subpanel shows $\log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}}$. Blue points indicate sources with $S/N > 5$ in both $[\text{O II}]$ and $[\text{O III}]$. The dashed diagonal line in each upper panel marks equality between the $[\text{O II}]$ and FIR SFRs, while the dashed horizontal line in the lower subpanel marks zero difference, $\log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}} = 0$. The median offset and scatter are labeled in each panel.

As shown in Figure 2a, after subtracting the $0.109L_{[\text{O III}]}$, the $[\text{O II}]$ SFRs are mainly lower than the FIR SFRs. For the full sample, the median offset is $\log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}} = -0.20 \pm 0.72$ dex. For the subsample with both $S/N > 5$ $[\text{O II}]$ and $[\text{O III}]$, the offset remains similar, -0.17 ± 0.69 dex. This suggests that for our quasar sample, the subtracting term $0.109L_{[\text{O III}]}$ may oversubtract the AGN contribution of $[\text{O II}]$, although the large scatter indicates substantial object-to-object variation.

We therefore also compare the FIR SFRs with the $[\text{O II}]$ SFRs without subtracting $0.109L_{[\text{O III}]}$, as shown in Figure 2b. In this case, the median offset becomes 0.00 ± 0.69 dex for the full sample, and the $S/N > 5$ subsample shows a positive offset of 0.12 ± 0.66 dex. This indicates that using the total $[\text{O II}]$ luminosity without subtracting AGN narrow line contamination may lead to an overestimation of $[\text{O II}]$ SFR.

One possible explanation is that the $[\text{O III}]$ luminosity used for the subtraction is not totally from the AGN NLR. If part of the $[\text{O III}]$ emission is from the star-forming regions of host galaxies, then using the total $[\text{O III}]$ luminosity to estimate the AGN contribution to $[\text{O II}]$ would oversubtract the $[\text{O II}]$ luminosity. Another possible explanation is that the coefficient 0.109 may not be applied to all AGNs, especially quasars. The relative strengths of $[\text{O II}]$ and $[\text{O III}]$ in NLR can depend on ionization parameter, gas density, metallicity, and the shape of the ionizing continuum.

In addition, although we use SFR_{FIR} as the reference SFR, it can also bring uncertainties. The FIR SFR are derived from SED decomposition, so they can be affected by different cold dust templates, AGN component, and photometric data coverage, although these uncertainties are smaller than the large scatter between $[\text{O II}]$ SFR and FIR SFR. In this work,

our main goal is to use the comparison to test how well [O II] performs for the sample as a whole, not to obtain highly precise SFR for each individual quasar.

Systematic uncertainties in the extinction correction may also contribute to the discrepancy. As described above, we use two methods to correct the [O II] and [O III] luminosities for dust attenuation: one is based on the Balmer decrement and the other is based on the attenuation from SED fitting. These two methods may introduce systematic differences in [O II] SFR results. We therefore compare the [O II] SFRs with the FIR SFRs separately for the two extinction correction methods in Figure 3. Blue points represent sources corrected using the Balmer decrement, while red points represent sources corrected using the SED attenuation.

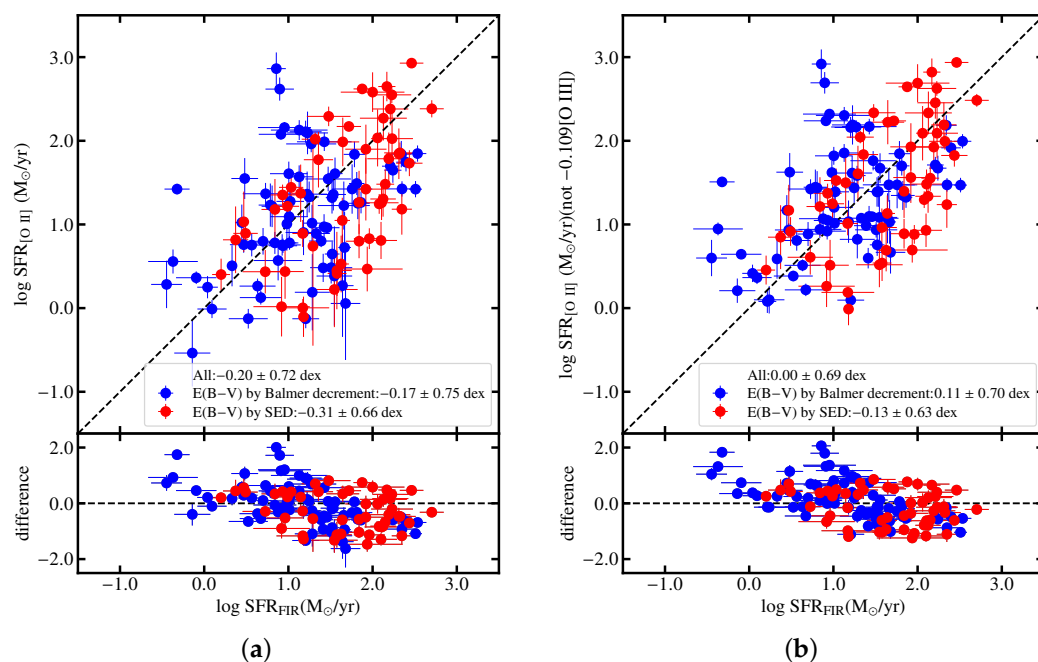


Figure 3. Comparison between [O II] $\lambda 3727$ SFRs and FIR SFRs using different extinction-correction methods. The axes and the dashed reference lines are the same as in Figure 2. Blue and red points denote sources corrected using the Balmer decrement and SED-fitting attenuation, respectively. Panel (a) shows the [O II] SFR after subtracting the AGN narrow-line contribution estimated as $0.109L_{[\text{O III}]}$, while panel (b) shows the [O II] SFR computed without this correction.

Figure 3 shows that, regardless of whether the $0.109L_{[\text{O III}]}$ subtraction is applied, the sources corrected using SED attenuation tend to have lower [O II] SFR than those corrected using the Balmer decrement. This suggests that extinction correction is an important source of systematic uncertainty in the [O II] SFR.

In addition, the comparison between [O II] and FIR SFR may be affected by aperture effects. The optical spectra sample the emission-line region within the fiber or slit aperture, whereas the FIR SFR is derived from the integrated dust emission of the host galaxy. Therefore, spatially extended star formation outside the spectroscopic aperture may cause the [O II] SFR to be lower than the FIR SFR. [O II] and FIR also trace different physical components and timescales, [O II] traces unobscured or moderately obscured recent star formation in ionized gas, while FIR traces dust-obscured star formation averaged over longer timescales. These effects, together with metallicity uncertainties and AGN contamination, likely contribute to the large scatter between the two SFR tracers.

Overall, [O II] can provide a useful but uncertain SFR tracer for quasar host galaxies, and its application requires careful treatment of AGN contamination, extinction correction, metallicity, and aperture effects.

As a further check, since the offset $\log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}}$ increases when a higher signal-to-noise ratio threshold is applied to the [O II] and [O III] emission lines, we further raise the threshold to $S/N > 10$ and examine how the offset changes. The results are shown in Figure 4. For the $S/N > 10$ subsample, after subtracting the AGN narrow-line contribution estimated as $0.109L_{[\text{O III}]}$, the median offset is $\log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}} = 0.05 \pm 0.69$ dex. If this narrow-line correction is not applied, the median offset becomes 0.26 ± 0.67 dex.

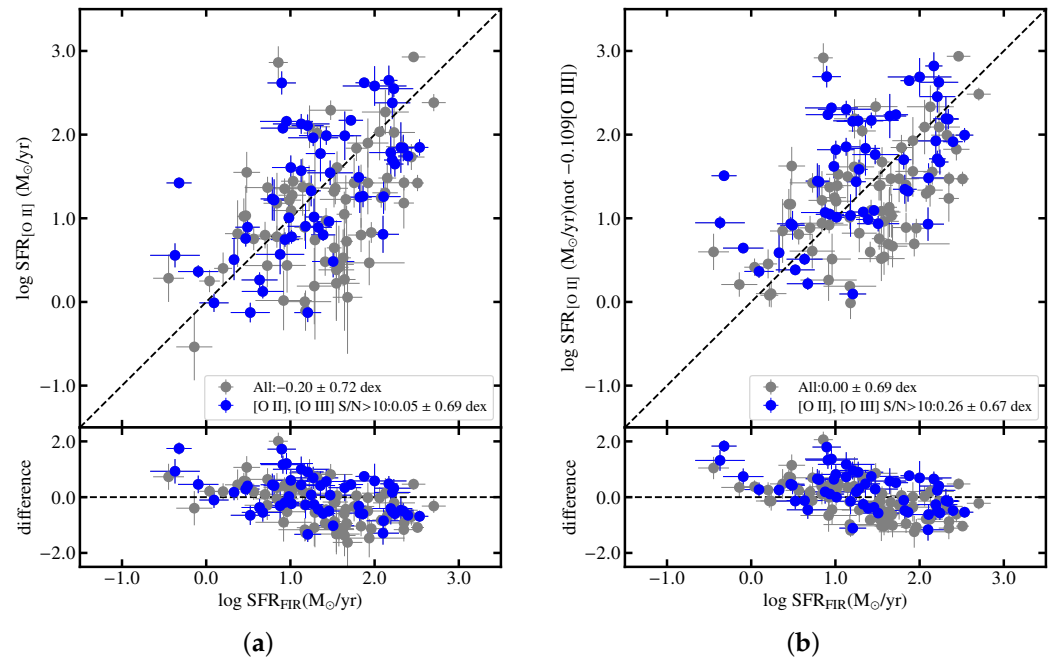


Figure 4. Comparison between [O II] $\lambda 3727$ SFRs and FIR SFRs for sources with $S/N > 10$ in both [O II] and [O III]. The axes, symbols and the dashed reference lines are the same as in Figure 2. Panel (a) shows the [O II] SFR after subtracting the AGN narrow-line contribution estimated as $0.109L_{[\text{O III}]}$, while panel (b) shows the [O II] SFR computed without this correction.

For this high- S/N subsample, Figure 4a shows that the [O II] SFRs corrected using the $0.109L_{[\text{O III}]}$ term are, on average, comparable to the FIR SFRs. However, this result should be interpreted with caution. A strict S/N selection prefers to select sources with stronger [O II] emission, and therefore may bias the sample toward objects with higher [O II] SFRs. Even for this high- S/N subsample, the corrected [O II] SFRs are only comparable to, rather than significantly higher than, the FIR SFRs, suggesting that the corrected [O II] SFRs may still be underestimated for some sources. The right panel shows that, if no correction for the AGN narrow-line contribution is applied, the [O II] SFRs are systematically higher than the FIR SFRs by 0.26 dex. This indicates that the AGN contribution to [O II] is non-negligible and should be accounted for when using [O II] as an SFR tracer in quasars. Overall, our comparison suggests that subtracting an AGN narrow-line contribution is necessary, but the simple correction term $0.109L_{[\text{O III}]}$ may oversubtract the [O II] luminosity for part of our sample.

Overall, the high- S/N comparison confirms that the AGN contribution to [O II] cannot be ignored. Without the $0.109L_{[\text{O III}]}$ subtraction, the [O II] SFRs are systematically higher than the FIR SFRs. However, the corrected [O II] SFRs are only comparable to the FIR SFRs even for the high- S/N subsample, suggesting that this correction may oversubtract the star formation-related [O II] emission in some quasars.

4.2. Dependence of the SFR Offset on AGN Activity

Since the observed [O II] $\lambda 3727$ emission can originate from both star-forming regions in the host galaxy and the AGN narrow-line region, the difference between the [O II] and FIR SFRs may depend on AGN activity. We therefore compare the SFR offset,

$$\Delta \log \text{SFR} = \log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}}, \quad (5)$$

with the bolometric luminosity, L_{bol} , and the Eddington ratio, $\lambda_{\text{Edd}} = L_{\text{bol}}/L_{\text{Edd}}$. The Eddington luminosity is given by

$$L_{\text{Edd}} = 1.26 \times 10^{38} \left(\frac{M_{\text{BH}}}{M_{\odot}} \right) \text{ erg s}^{-1}. \quad (6)$$

The results are shown in Figure 5. The left panel compares $\Delta \log \text{SFR}$ with L_{bol} , and the right panel compares $\Delta \log \text{SFR}$ with λ_{Edd} . The Spearman correlation coefficient, ρ , and the corresponding p value are shown in the lower right corner of each panel. For both the full sample and the subsample with $\text{S/N} > 10$ [O II] and [O III], there is no statistically significant correlation between $\Delta \log \text{SFR}$ and L_{bol} , although a weak negative trend is visually suggested. For λ_{Edd} , the full sample shows only a very weak negative trend, and no significant correlation is found for the high S/N subsample.

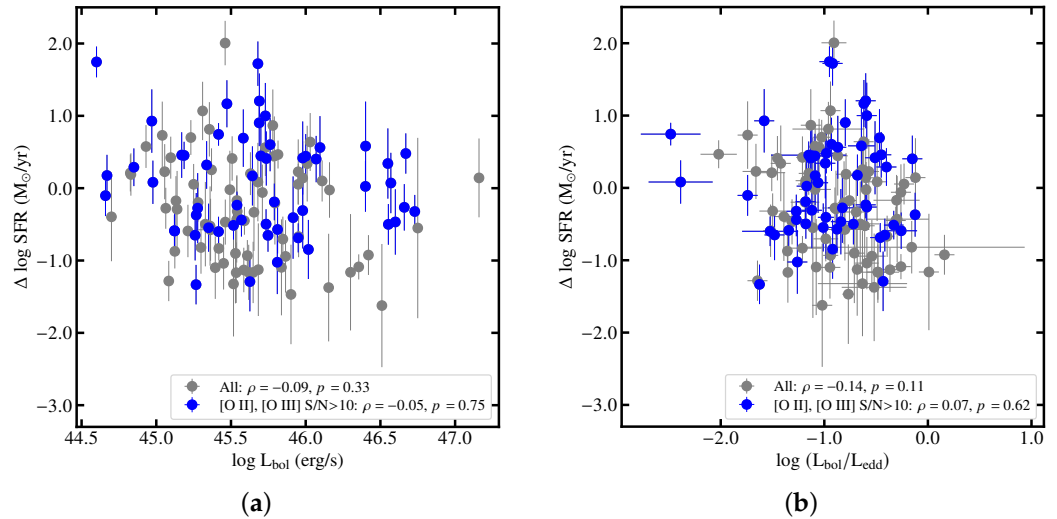


Figure 5. SFR offset, $\Delta \log \text{SFR} = \log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}}$, as a function of (a) bolometric luminosity and (b) Eddington ratio. The Spearman correlation coefficient, ρ , and the corresponding p value are shown in the lower right corner of each panel for both the full sample and the subsample with $\text{S/N} > 10$ in both [O II] and [O III].

We also compare $\Delta \log \text{SFR}$ with the fractional AGN contribution to the total infrared luminosity, f_{AGN} (8–1000 μm), derived from the SED fitting. The result is shown in Figure 6. For the full sample, the Spearman correlation coefficient is $\rho = 0.34$, with $p < 10^{-3}$, indicating a weak positive correlation. For the high-S/N subsample, the correlation becomes somewhat stronger, with $\rho = 0.47$ and $p < 10^{-3}$. To illustrate the overall trend, we divide the sample into six bins in f_{AGN} (8–1000 μm) and show the median value in each bin as black squares.

For sources with f_{AGN} (8–1000 μm) $\lesssim 0.6$, $\Delta \log \text{SFR}$ shows no clear dependence on f_{AGN} , and the [O II] SFRs tend to be lower than the FIR SFRs. At higher f_{AGN} , however, $\Delta \log \text{SFR}$ increases with increasing f_{AGN} and can become positive, indicating that the [O II] SFRs exceed the FIR SFRs for sources with large AGN infrared contributions. This behavior may have several possible origins. First, a larger f_{AGN} (8–1000 μm) implies a smaller relative contribution from the cold dust component used to derive SFR_{FIR} , which

can lower the FIR SFR. Second, when the AGN contribution is weak, a non-negligible fraction of the observed [O III] emission may be associated with star-forming regions in the host galaxy. In this case, using the total $L_{[\text{O III}]}$ to estimate the AGN narrow-line contribution to [O II] may over-subtract the star formation-related [O II] emission, leading to underestimated [O II] SFRs.

However, the correlation with f_{AGN} should be interpreted with caution. Both f_{AGN} and SFR_{FIR} are derived from the same SED decomposition, so part of the observed trend may reflect covariance between the AGN and cold-dust components in the SED fitting. In addition, uncertainties in dust attenuation, metallicity, aperture effects, and the different timescales traced by optical emission lines and FIR emission may also contribute to the scatter in $\Delta \log \text{SFR}$.

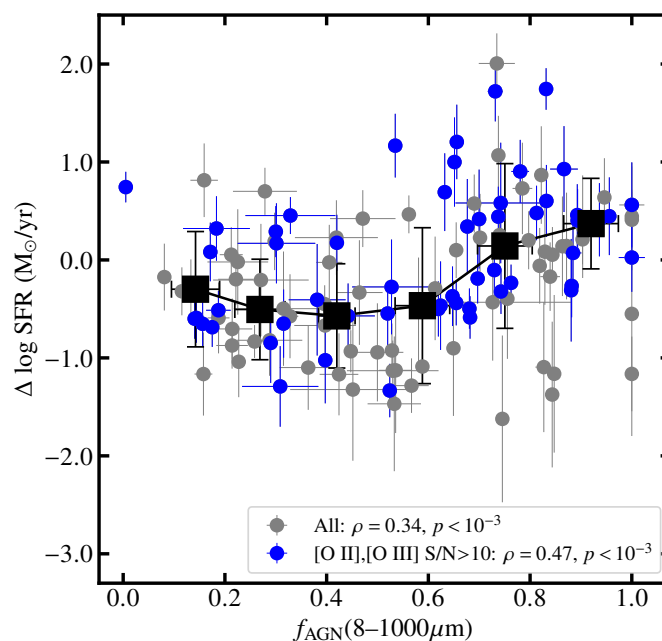


Figure 6. SFR offset, $\Delta \log \text{SFR} = \log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}}$, as a function of the fractional AGN contribution to the total infrared luminosity, $f_{\text{AGN}}(8\text{--}1000 \mu\text{m})$. The Spearman correlation coefficient, ρ , and the corresponding p value are shown in the lower right corner for both the full sample and the subsample with $\text{S/N} > 10$ in both [O II] and [O III]. Black squares show the median values in six bins of f_{AGN} .

4.3. Dependence of the SFR Offset on Redshift

We further examine whether the SFR offset ($\Delta \log \text{SFR}$) depends on redshift. The result is shown in Figure 7. For the full sample, the Spearman correlation coefficient is $\rho = -0.32$, with $p < 10^{-3}$, indicating a weak negative correlation between $\Delta \log \text{SFR}$ and redshift. To better illustrate the overall trend, we divide the sample into five redshift bins and show the median value in each bin as black squares.

For quasars at $z < 0.4$, $\Delta \log \text{SFR}$ decreases gradually with increasing redshift. At $z > 0.4$, the median offset becomes relatively flat, but the [O II] SFRs are generally lower than the FIR SFRs. In particular, the offset is close to zero for the low-redshift subsample at $z < 0.2$, while the [O II] SFRs tend to be underestimated relative to the FIR SFRs for sources at higher redshift. This suggests that, for quasars at $z \gtrsim 0.4\text{--}0.5$, the [O II] SFRs derived using Equation (4) should be interpreted with caution. This redshift dependence may partly arise from methodological effects, including the lack of $\text{H}\alpha$ coverage at higher redshift, increased reliance on SED attenuation corrections, aperture effects, FIR selection bias, and uncertainties in the AGN NLR correction, rather than pure physical evolution.

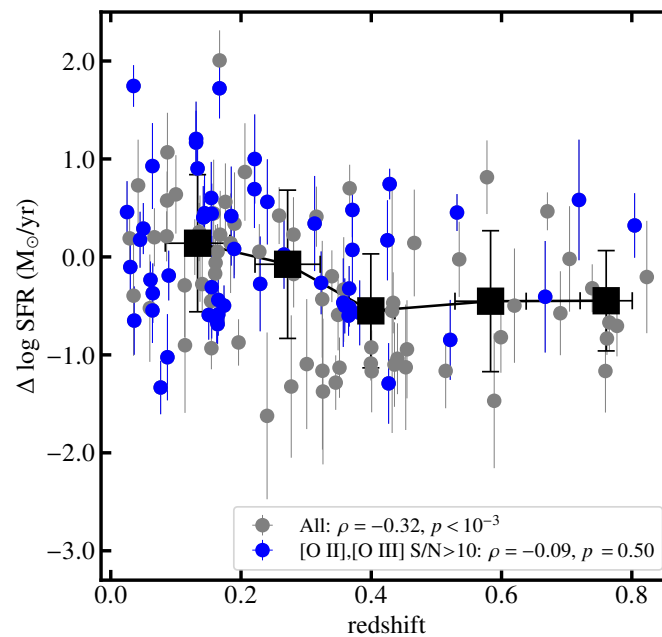


Figure 7. SFR offset, $\Delta \log \text{SFR}$, as a function of redshift. The Spearman correlation coefficient, ρ , and the corresponding p value are shown in the lower right corner for both the full sample and the subsample with $S/N > 10$ in both [O II] and [O III]. Black squares show the median values in five redshift bins.

4.4. Sensitivity to the Adopted AGN NLR Correction

The comparisons above show that neither ignoring the AGN contribution nor applying a fixed $0.109L_{[\text{O III}]}$ subtraction can provide an equally accurate [O II] SFR for every quasar. If no AGN correction is applied, the observed [O II] emission may still include a contribution from the AGN NLR. On the other hand, using the same $0.109L_{[\text{O III}]}$ correction for all sources may remove too much or too little AGN-related [O II] emission in some objects. We therefore compare the SFR offset ($\Delta \log \text{SFR} = \log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}}$) calculated with and without subtracting $0.109L_{[\text{O III}]}$. We examine how this offset changes with AGN luminosity, Eddington ratio, infrared AGN fraction, and redshift.

Figure 8 shows no statistically significant dependence of the SFR offset on L_{bol} for either treatment. The corresponding Spearman coefficients are $\rho = -0.09$ ($p = 0.33$) after the subtraction and $\rho = -0.03$ ($p = 0.76$) without the subtraction. Similarly, the dependence on λ_{Edd} is weak. The uncorrected result shows a little negative correlation ($\rho = -0.17$, $p = 0.05$), while the corrected result is not statistically significant ($\rho = -0.14$, $p = 0.11$). So, within the dynamic range of our sample, we find no strong evidence that the relative performance of the fixed correction depends systematically on AGN luminosity or accretion state. As for the infrared AGN fraction, the SFR offset increases with f_{AGN} both before and after the subtraction, but the correlation becomes weaker after the $0.109L_{[\text{O III}]}$ term is applied, decreasing from $\rho = 0.45$ ($p < 10^{-6}$) to $\rho = 0.34$ ($p < 10^{-3}$). This suggests that the adopted correction removes part of the AGN-related excess in the observed [O II] emission. However, the remaining positive trend indicates that a single [O II]/[O III] ratio does not fully describe the object-to-object diversity of the quasar NLR. However, this trend should be interpreted with caution because f_{AGN} and SFR_{FIR} are both derived from the same SED decomposition and are therefore not completely independent. The SFR offset decreases with redshift both with and without the AGN correction, with $\rho = -0.32$ ($p < 10^{-3}$) and $\rho = -0.35$ ($p < 10^{-4}$), respectively. Since the redshift trend is present with comparable strength in both cases, it is unlikely to be caused primarily by the fixed $0.109L_{[\text{O III}]}$ subtraction. Instead, it may reflect a combination of aperture effects, changes in

the extinction-correction method when narrow H α is unavailable, sample selection, and the different physical timescales traced by optical emission lines and FIR emission.

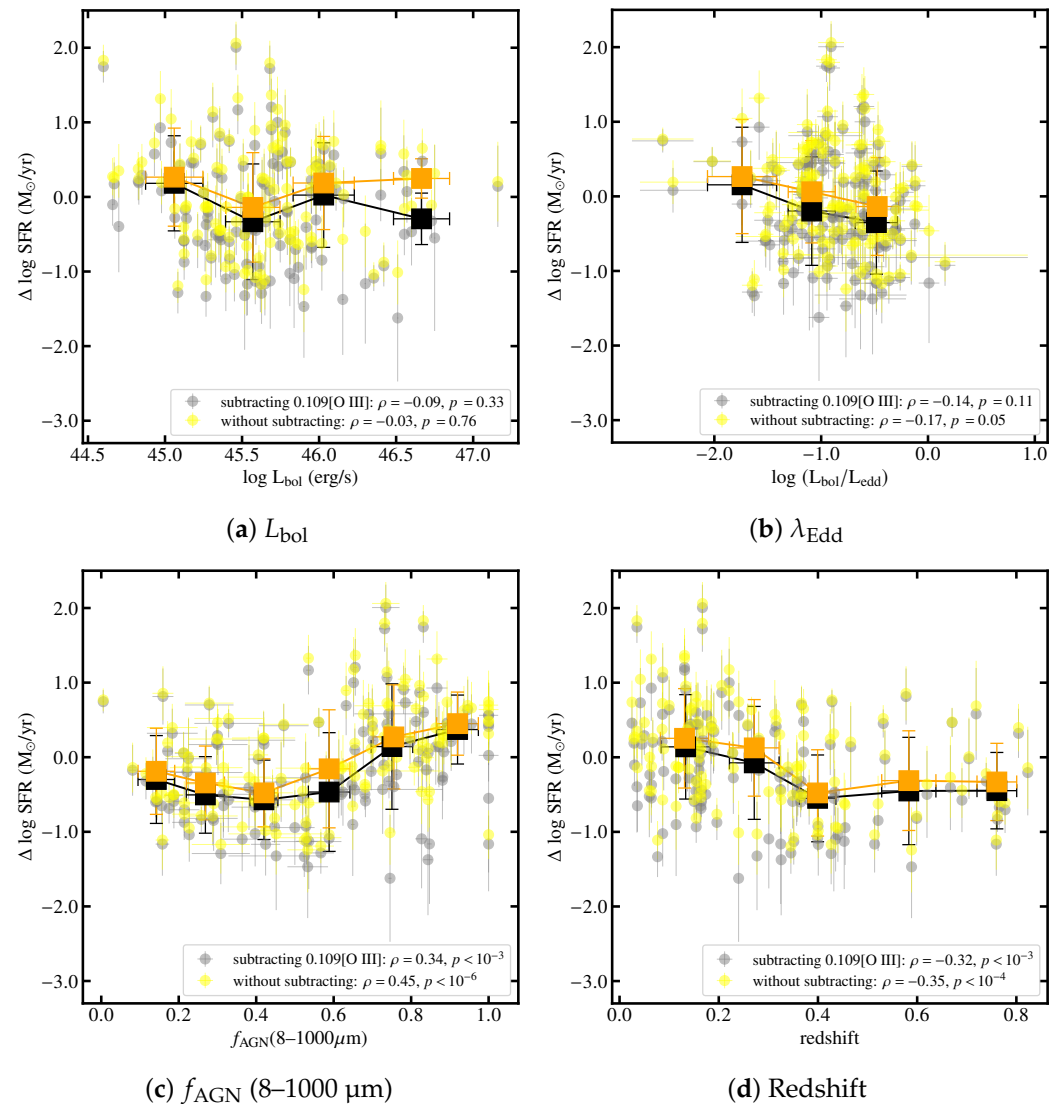


Figure 8. Comparison of the SFR offset, $\Delta \log \text{SFR} = \log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}}$, derived with and without subtracting the AGN narrow-line contribution $0.109L_{[\text{O III}]}$. The panels show the dependence on (a) bolometric luminosity, (b) Eddington ratio, (c) infrared AGN fraction and (d) redshift. The two sets of symbols correspond to the results with and without the $0.109L_{[\text{O III}]}$ subtraction, as indicated in each panel. Black and orange squares show the median values in bins for the results with and without the $0.109L_{[\text{O III}]}$ subtraction, respectively.

Overall, the comparison indicates that the AGN contribution to [O II] cannot be ignored statistically, but that the fixed $0.109L_{[\text{O III}]}$ subtraction should not be interpreted as an exact object-by-object correction. It reduces part of the AGN-related bias, particularly for sources with a large infrared AGN contribution, while substantial residual scatter remains.

A recent study by Wu et al. [22] used a kinematically consistent decomposition of [O II] $\lambda 3727$ and the high-ionization [Ne V] $\lambda 3426$ line to estimate the AGN contribution to [O II] emission in SDSS quasars. They found that the narrow [O II] emission is dominated by star formation on average, with an estimated AGN contribution of only $\sim 6\%$ in their composite spectrum. When weak broad and very broad [O II] components were included, the inferred AGN contribution increased to $\sim 17\%$. Their result supports the use of [O II] as a statistical SFR tracer in quasars. However, they also noted that the AGN contribution can

vary among individual sources and that reliable measurements of the [O II] and [Ne V] line profiles are needed for an object-by-object correction. Our results are broadly consistent with this picture. The comparison with FIR SFRs indicates that AGN contamination in [O II] cannot be ignored, especially for sources with strong AGN-related emission. At the same time, the systematic offset introduced by the fixed $0.109L_{[\text{O III}]}$ subtraction suggests that a universal [O II]/[O III] ratio may not provide an accurate correction for every quasar. A line-profile-based method using [Ne V], such as that of Wu et al. [22], may provide a useful alternative when spectra of sufficient quality are available.

5. Summary and Conclusions

We select a sample of 202 quasars, including 120 SDSS quasars and 87 PG quasars, with five overlapping objects. We measure [O II] $\lambda 3727$, [O III] $\lambda 5007$, and narrow Balmer emission lines by fitting SDSS spectra and newly observed optical spectra of PG quasars. Then, we investigate the reliability of the [O II] $\lambda 3727$ emission line as an SFR tracer for the quasar host galaxies by comparing [O II] SFRs with FIR SFRs. The FIR SFRs are derived from multiwavelength SED decomposition and are used as an independent SFR reference. Our main results are summarized as follows.

1. We refit the SDSS spectra of 120 SDSS quasars and 31 PG quasars with available SDSS spectra using QSOFITMORE. For the remaining PG quasars without suitable archival spectra, we obtained new optical spectroscopic observations with XLT, LJT, HCT, and P200/DBSP. We got new spectra for 52 PG quasars. These spectra-fitting results provide the narrow emission line luminosities for calculating [O II] SFRs.
2. We calculate [O II] SFRs using the metallicity-dependent calibration of Zhuang and Ho [4]. We subtract the AGN NLR contribution to [O II], estimated as $0.109L_{[\text{O III}]}$. Extinction is corrected using the Balmer decrement when reliable narrow H α and H β measurements are available. Otherwise, we use the extinction obtained from SED fitting. The gas metallicity is estimated from the stellar mass–metallicity relation.
3. After subtracting the $0.109L_{[\text{O III}]}$ contribution, [O II] SFRs are mainly lower than FIR SFRs. The median offset is $\log \text{SFR}_{[\text{O II}]} - \log \text{SFR}_{\text{FIR}} = -0.20 \pm 0.72$ dex for the full sample and -0.17 ± 0.69 dex for sources with $\text{S/N} > 5$ [O II] and [O III]. This suggests that the $0.109L_{[\text{O III}]}$ correction may oversubtract the star formation-related [O II] luminosity for some quasars, although the large scatter indicates substantial object-to-object variation.
4. If the AGN NLR subtraction is not applied, the median offset becomes 0.00 ± 0.69 dex for the full sample and 0.12 ± 0.66 dex for the $\text{S/N} > 5$ subsample. For the more strict $\text{S/N} > 10$ subsample, the corrected [O II] SFRs are comparable to FIR SFRs, with a median offset of 0.05 ± 0.69 dex, while the uncorrected [O II] SFRs are higher by 0.26 ± 0.67 dex. These results show that the AGN contribution to [O II] is non-negligible and should be considered when using [O II] as an SFR tracer of quasar host galaxies.
5. The extinction correction is an important source of systematic uncertainty. Sources corrected using the SED extinction tend to have lower [O II] SFRs than sources corrected using the Balmer decrement. This indicates that the uncertainty of the narrow-line gas extinction can contribute significantly to the scatter between [O II] and FIR SFRs.
6. We find no statistically significant correlation between the SFR offset and bolometric luminosity or Eddington ratio. However, the SFR offset shows a weak positive correlation with the fractional AGN contribution to the total infrared luminosity, f_{AGN} (8–1000 μm), with $\rho = 0.34$ and $p < 10^{-3}$ for the full sample. The correlation is stronger for the high S/N subsample. This trend may reflect the combined effects

of AGN contamination, SED decomposition component degeneracy, and the relative contributions of AGN-heated dust and cold dust.

7. The SFR offset shows a weak negative correlation with redshift for the full sample, with $\rho = -0.32$ and $p < 10^{-3}$. The [O II] SFRs are close to the FIR SFRs at $z < 0.2$, but tend to be lower than the FIR SFRs at $z \gtrsim 0.4$ – 0.5 . This redshift dependence may partly arise from methodological effects, including the lack of H α coverage at higher redshift, increased reliance on SED extinction corrections, aperture effects, FIR selection bias, and uncertainties in the AGN NLR correction.

Overall, our results show that [O II] $\lambda 3727$ is useful for studying the statistical SFR properties of quasar host galaxies, but it has large uncertainties for individual objects. An AGN NLR correction is needed to reduce AGN contamination. Otherwise, the [O II] SFR may be overestimated. In addition, a fixed subtraction of $0.109L_{[\text{O III}]}$ may oversubtract the [O II] luminosity of some quasars. Therefore, reliable use of [O II] requires careful consideration of AGN contamination, dust extinction, metallicity, aperture effects, and redshift-related biases.

Author Contributions: Conceptualization, X.F. and X.-B.W.; methodology, X.F., X.-B.W. and Y.F.; software, X.F. and Y.F.; validation, X.F. and Y.F.; formal analysis, X.F.; investigation, X.F., X.-B.W., Y.F., Y.P., R.Z. and H.W.; resources, X.F., X.-B.W., Y.F. and Y.P.; data curation, X.F., X.-B.W., Y.F., Y.P., R.Z. and H.W.; writing—original draft preparation, X.F.; writing—review and editing, X.F., X.-B.W., Y.F., Y.P., R.Z. and H.W.; visualization, X.F.; supervision, X.-B.W.; project administration, X.-B.W.; funding acquisition, X.-B.W. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The public data used in this study are available from the Sloan Digital Sky Survey, the NASA/IPAC Infrared Science Archive, the NASA/IPAC Extragalactic Database, the Herschel Science Archive, and Pan-STARRS1. The newly obtained optical spectra and derived measurements presented in this work are available from the corresponding author upon reasonable request.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AGN	Active galactic nucleus
BC	Balmer continuum
DBSP	Double Spectrograph
FIR	Far-infrared
FWHM	Full width at half maximum
HCT	Himalayan Chandra Telescope
IRSA	NASA/IPAC Infrared Science Archive
JCMT	James Clerk Maxwell Telescope
LJT	Lijiang 2.4 m Telescope
NED	NASA/IPAC Extragalactic Database
NLR	Narrow-line region
PACS	Photodetector Array Camera and Spectrometer
PCA	Principal component analysis
PG	Palomar-Green
SDSS	Sloan Digital Sky Survey
SED	Spectral energy distribution
SFR	Star formation rate
S/N	Signal-to-noise ratio
SPIRE	Spectral and Photometric Imaging Receiver
XLT	Xinglong 2.16 m Telescope

Appendix A. Emission-Line and Star Formation Rate Measurements

This appendix provides the object-by-object measurements used in the analysis. Appendix Tables A1–A3 list the narrow emission-line luminosities measured from the SDSS and newly obtained optical spectra. Appendix A Table A4 lists the extinction corrections, corrected line luminosities, FIR reference SFRs, and [O II] SFRs used for the comparison in Section 4.

Table A1. Emission-line measurements for the SDSS quasar sample.

No.	Name	z	$\log L_{[\text{O II}]}$ (erg s^{-1})	$\log L_{[\text{O III}]}$ (erg s^{-1})	$\log L_{\text{H}\alpha}$ (erg s^{-1})	$\log L_{\text{H}\beta}$ (erg s^{-1})
1	J001115.69+011459.23	0.580	41.60 ± 0.10	41.28 ± 0.36	–	41.51 ± 0.08
2	J002421.84+002508.49	0.599	41.39 ± 0.10	41.50 ± 0.08	–	41.81 ± 0.05
3	J005905.51+000651.67	0.719	42.46 ± 0.02	43.11 ± 0.02	–	42.50 ± 0.02
4	J011430.24+000420.94	0.455	41.05 ± 0.09	40.26 ± 1.40	–	39.89 ± 0.84
5	J011536.92+000011.01	0.532	41.67 ± 0.03	42.21 ± 0.02	–	41.51 ± 0.06
6	J013354.30+011034.00	0.704	41.78 ± 0.05	41.86 ± 0.11	–	41.48 ± 0.35
7	J014003.99+002541.77	0.770	42.08 ± 0.03	41.89 ± 0.27	–	41.46 ± 0.25
8	J015243.15+002039.65	0.578	41.78 ± 0.07	42.14 ± 0.06	–	41.15 ± 0.26
9	J015950.25+002340.82	0.163	41.35 ± 0.02	42.26 ± 0.01	42.54 ± 0.01	41.59 ± 0.02
10	J021859.87+002855.80	0.351	41.36 ± 0.06	41.55 ± 0.04	42.36 ± 0.03	41.87 ± 0.03
11	J022430.61+000038.91	0.431	41.44 ± 0.03	41.42 ± 0.20	–	41.78 ± 0.07
12	J090026.51+204158.70	0.706	41.96 ± 0.09	42.16 ± 1.65	–	–
13	J090158.88+002313.87	0.196	41.23 ± 0.02	41.74 ± 0.04	41.76 ± 0.01	41.06 ± 0.03
14	J090933.49+425346.51	0.670	42.65 ± 0.01	42.25 ± 0.13	–	41.62 ± 0.22
15	J092159.40+450912.38	0.235	41.88 ± 0.01	41.20 ± 0.20	42.30 ± 0.02	41.55 ± 0.01
16	J092554.72+195405.13	0.192	40.76 ± 0.23	42.09 ± 0.01	42.26 ± 0.03	41.84 ± 0.03
17	J092635.12+072446.43	0.189	41.44 ± 0.01	42.03 ± 0.01	41.70 ± 0.02	40.91 ± 0.04
18	J094652.58+131953.83	0.133	42.12 ± 0.01	42.44 ± 0.04	42.83 ± 0.01	42.01 ± 0.03
19	J094745.14+072520.58	0.086	40.96 ± 0.02	41.56 ± 0.05	41.66 ± 0.01	41.02 ± 0.02

Table A1. Cont.

No.	Name	<i>z</i>	$\log L_{[\text{O II}]}$ (erg s^{-1})	$\log L_{[\text{O III}]}$ (erg s^{-1})	$\log L_{\text{H}\beta}$ (erg s^{-1})	$\log L_{\text{H}\alpha}$ (erg s^{-1})
20	J095017.06+215022.41	0.455	40.91 ± 0.14	41.91 ± 0.02	42.61 ± 0.01	41.77 ± 0.03
21	J095240.17+515249.91	0.554	41.24 ± 0.13	42.25 ± 0.02	–	40.70 ± 0.58
22	J095819.87+022903.51	0.345	40.80 ± 0.11	41.14 ± 0.10	41.49 ± 0.10	40.75 ± 0.10
23	J100043.14+020637.25	0.360	41.20 ± 0.04	41.38 ± 0.15	41.90 ± 0.03	40.70 ± 0.08
24	J100232.13+023537.33	0.658	41.60 ± 0.06	42.22 ± 0.05	–	41.89 ± 0.12
25	J100420.13+051300.48	0.160	41.10 ± 0.05	41.81 ± 0.01	42.80 ± 0.01	42.05 ± 0.01
26	J103651.94+575950.96	0.500	41.54 ± 0.06	41.65 ± 0.23	–	41.56 ± 0.12
27	J104009.33+560343.28	0.393	41.44 ± 0.05	42.07 ± 0.03	42.56 ± 0.04	41.88 ± 0.05
28	J104505.39+561118.34	0.428	42.36 ± 0.01	42.43 ± 0.04	–	41.87 ± 0.02
29	J104739.49+563507.19	0.303	40.98 ± 0.08	41.18 ± 0.18	41.58 ± 0.11	40.96 ± 0.08
30	J105106.12+591625.24	0.767	42.08 ± 0.09	42.45 ± 0.15	–	–
31	J105143.89+335926.71	0.167	41.33 ± 0.04	42.06 ± 0.10	42.45 ± 0.01	41.23 ± 0.04
32	J105151.44+005117.66	0.359	42.46 ± 0.02	43.37 ± 0.01	43.02 ± 0.01	42.31 ± 0.02
33	J105705.41+580437.46	0.140	40.62 ± 0.05	41.35 ± 0.02	41.48 ± 0.02	40.60 ± 0.13
34	J105959.93+574848.17	0.453	41.36 ± 0.06	42.38 ± 0.04	–	41.72 ± 0.04
35	J110036.64+564134.89	0.834	41.58 ± 0.12	–	–	–
36	J111706.40+441333.30	0.144	41.41 ± 0.04	42.25 ± 0.03	42.44 ± 0.01	41.65 ± 0.04
37	J111830.29+402554.02	0.154	41.13 ± 0.06	41.63 ± 0.09	42.84 ± 0.01	42.17 ± 0.02
38	J112019.62+130320.10	0.314	41.28 ± 0.04	41.54 ± 0.23	42.34 ± 0.01	41.13 ± 0.34
39	J112048.99+133821.91	0.513	41.52 ± 0.05	41.83 ± 0.07	–	41.38 ± 0.06
40	J112759.26+360207.00	0.667	41.84 ± 0.03	42.55 ± 0.03	–	41.45 ± 0.11
41	J114121.76+014803.58	0.382	41.62 ± 0.04	42.09 ± 0.03	42.46 ± 0.02	41.73 ± 0.03
42	J114933.88+222227.07	0.554	41.32 ± 0.09	41.27 ± 0.35	–	41.30 ± 0.11
43	J115253.68+000131.90	0.823	41.91 ± 0.07	42.35 ± 0.07	–	41.51 ± 0.50
44	J120226.76+012915.28	0.150	41.25 ± 0.02	41.36 ± 0.01	42.02 ± 0.01	41.10 ± 0.03
45	J120312.14+015321.30	0.296	41.02 ± 0.04	40.80 ± 0.25	41.67 ± 0.02	41.54 ± 0.07
46	J120442.11+275411.80	0.165	41.69 ± 0.01	42.32 ± 0.01	42.22 ± 0.01	41.59 ± 0.01
47	J120734.63+150643.69	0.750	41.74 ± 0.08	42.56 ± 0.18	–	42.54 ± 0.12
48	J121037.81+053805.88	0.436	41.09 ± 0.11	41.75 ± 0.08	–	41.44 ± 0.12
49	J121728.37+065110.00	0.589	41.56 ± 0.09	42.19 ± 0.08	–	41.55 ± 0.15
50	J121836.71+155908.43	0.766	42.01 ± 0.05	42.33 ± 0.10	–	42.45 ± 0.07
51	J121906.57+160243.06	0.761	42.11 ± 0.03	41.55 ± 0.56	–	41.50 ± 0.34
52	J121945.03+082117.95	0.228	41.18 ± 0.02	40.86 ± 0.10	42.05 ± 0.01	41.07 ± 0.03
53	J121946.54+145259.37	0.401	41.38 ± 0.05	41.90 ± 0.05	41.14 ± 0.64	40.94 ± 0.15
54	J122011.88+020342.21	0.240	41.95 ± 0.02	42.79 ± 0.01	42.89 ± 0.01	42.04 ± 0.02
55	J122026.72+062748.20	0.349	40.67 ± 0.12	41.27 ± 0.14	40.82 ± 0.32	40.44 ± 0.16
56	J122102.50+155447.04	0.229	41.17 ± 0.04	41.81 ± 0.01	42.08 ± 0.03	41.30 ± 0.05
57	J122102.95+000733.74	0.366	41.94 ± 0.01	41.64 ± 0.03	42.69 ± 0.02	41.96 ± 0.01
58	J122106.50+114625.46	0.340	41.40 ± 0.03	41.80 ± 0.06	42.10 ± 0.02	41.31 ± 0.03
59	J122307.37+002124.73	0.804	42.09 ± 0.03	42.68 ± 0.02	–	41.40 ± 0.34
60	J122312.17+095017.72	0.277	40.96 ± 0.08	41.80 ± 0.10	41.83 ± 0.11	41.40 ± 0.10
61	J122317.80+092306.94	0.682	41.94 ± 0.10	41.86 ± 0.15	–	42.48 ± 0.04
62	J122404.62+045637.94	0.358	41.80 ± 0.02	42.01 ± 0.11	42.44 ± 0.04	41.72 ± 0.06
63	J122520.13+084450.76	0.535	41.94 ± 0.06	42.31 ± 0.05	–	42.02 ± 0.07
64	J122526.21+141332.24	0.760	41.76 ± 0.09	41.94 ± 0.11	–	42.19 ± 0.07
65	J122641.50+055906.81	0.290	41.11 ± 0.06	41.74 ± 0.16	41.78 ± 0.02	40.99 ± 0.07
66	J122822.10+114606.83	0.365	41.71 ± 0.02	41.98 ± 0.01	42.24 ± 0.02	41.56 ± 0.02
67	J122839.20+035749.29	0.608	41.35 ± 0.07	41.57 ± 0.36	–	40.94 ± 0.47
68	J123436.54+123918.66	0.777	42.10 ± 0.02	42.56 ± 0.07	–	41.67 ± 0.19
69	J123800.92+621336.09	0.440	41.15 ± 0.09	41.61 ± 0.03	–	41.65 ± 0.06
70	J124511.26+335610.12	0.711	–	42.41 ± 0.02	–	42.36 ± 0.05
71	J125257.36+331555.81	0.367	41.50 ± 0.03	41.63 ± 0.06	41.84 ± 0.11	41.09 ± 0.09
72	J125317.57+310550.64	0.782	42.29 ± 0.05	42.90 ± 0.08	–	42.41 ± 0.14

Table A1. Cont.

No.	Name	z	$\log L_{[\text{O II}]}$ (erg s^{-1})	$\log L_{[\text{O III}]}$ (erg s^{-1})	$\log L_{\text{H}\beta}$ (erg s^{-1})	$\log L_{\text{H}\alpha}$ (erg s^{-1})
73	J125553.05+272405.23	0.316	41.04 ± 0.07	41.65 ± 0.03	41.65 ± 0.07	40.87 ± 0.12
74	J125703.80+250457.56	0.821	41.65 ± 0.09	—	—	—
75	J125711.97+274216.45	0.793	41.76 ± 0.04	41.91 ± 0.36	—	41.45 ± 0.39
76	J125757.23+322929.29	0.806	41.39 ± 0.22	42.44 ± 0.03	—	41.72 ± 0.22
77	J130622.96+225752.95	0.758	41.70 ± 0.06	40.94 ± 1.14	—	41.30 ± 0.32
78	J130947.00+081948.23	0.154	41.71 ± 0.03	42.57 ± 0.03	42.66 ± 0.01	41.84 ± 0.02
79	J131247.97+250756.71	0.425	42.11 ± 0.02	42.64 ± 0.01	—	41.87 ± 0.02
80	J131312.13+284730.01	0.259	40.83 ± 0.07	40.85 ± 0.12	—	40.31 ± 0.15
81	J131531.66+265414.73	0.620	41.59 ± 0.08	42.28 ± 0.11	—	41.50 ± 0.17
82	J132919.84+250626.45	0.781	41.39 ± 0.13	41.58 ± 0.25	—	—
83	J133005.71+254243.77	0.598	—	42.34 ± 0.03	—	41.56 ± 0.13
84	J133442.29+320939.60	0.649	41.24 ± 0.20	41.51 ± 0.22	—	42.28 ± 0.06
85	J134356.74+253847.69	0.086	40.63 ± 0.05	41.49 ± 0.03	41.69 ± 0.01	40.74 ± 0.03
86	J135632.80+210352.35	0.301	41.58 ± 0.06	42.57 ± 0.01	42.43 ± 0.03	41.79 ± 0.03
87	J140621.89+222346.54	0.098	40.29 ± 0.09	—	42.51 ± 0.00	41.93 ± 0.00
88	J140655.66+015712.88	0.427	41.40 ± 0.04	41.93 ± 0.02	—	41.88 ± 0.05
89	J140700.40+282714.65	0.077	41.10 ± 0.02	41.75 ± 0.01	41.84 ± 0.01	41.26 ± 0.02
90	J141637.45+003352.28	0.434	41.83 ± 0.02	42.55 ± 0.04	—	41.74 ± 0.03
91	J141644.62+190541.92	0.365	41.27 ± 0.04	40.15 ± 1.17	41.97 ± 0.03	41.31 ± 0.14
92	J141700.83+445606.39	0.113	40.45 ± 0.07	41.25 ± 0.04	42.25 ± 0.01	—
93	J142052.44+525622.42	0.677	41.68 ± 0.08	42.50 ± 0.02	—	42.13 ± 0.04
94	J142648.78+005323.24	0.220	41.50 ± 0.01	41.92 ± 0.16	42.20 ± 0.01	41.21 ± 0.04
95	J142710.94+013023.22	0.704	42.30 ± 0.04	41.86 ± 0.26	—	41.91 ± 0.27
96	J142753.79+345248.35	0.514	41.26 ± 0.12	41.75 ± 0.04	—	42.18 ± 0.03
97	J142918.15+592106.65	0.739	42.24 ± 0.02	42.86 ± 0.08	—	42.16 ± 0.06
98	J142943.07+474726.23	0.221	41.53 ± 0.02	42.51 ± 0.01	42.91 ± 0.00	41.95 ± 0.01
99	J143624.81-002905.36	0.325	40.93 ± 0.10	41.98 ± 0.01	42.70 ± 0.01	41.73 ± 0.03
100	J144231.82+014353.43	0.280	40.79 ± 0.08	40.98 ± 0.12	41.71 ± 0.04	40.82 ± 0.14
101	J145001.69+022006.75	0.520	41.75 ± 0.04	—	—	41.81 ± 0.08
102	J145108.76+270926.92	0.064	41.02 ± 0.02	41.77 ± 0.01	42.50 ± 0.01	41.85 ± 0.01
103	J145538.73+002238.06	0.434	40.83 ± 0.16	41.61 ± 0.05	—	41.19 ± 0.09
104	J152114.26+222743.87	0.136	40.87 ± 0.05	41.62 ± 0.01	42.72 ± 0.01	41.78 ± 0.02
105	J154530.24+484608.98	0.399	41.68 ± 0.05	41.89 ± 0.03	43.48 ± 0.00	42.75 ± 0.01
106	J155444.58+082221.48	0.119	40.42 ± 0.10	41.56 ± 0.01	42.70 ± 0.00	42.00 ± 0.01
107	J161413.20+260416.21	0.131	42.24 ± 0.01	43.00 ± 0.02	42.83 ± 0.00	42.01 ± 0.01
108	J163352.34+402115.66	0.782	41.79 ± 0.05	42.12 ± 0.18	—	41.97 ± 0.12
109	J163915.81+412833.70	0.690	41.66 ± 0.13	42.26 ± 0.04	—	41.53 ± 0.15
110	J171033.22+584456.86	0.281	41.01 ± 0.05	41.25 ± 0.04	41.98 ± 0.01	41.09 ± 0.03
111	J171352.43+584201.25	0.521	42.03 ± 0.03	42.74 ± 0.02	—	41.63 ± 0.07
112	J220759.39+001722.62	0.368	40.84 ± 0.08	40.71 ± 0.20	42.75 ± 0.00	42.08 ± 0.01
113	J223607.68+134355.32	0.326	41.49 ± 0.05	42.58 ± 0.01	43.32 ± 0.00	42.62 ± 0.07
114	J233741.32+001743.77	0.762	41.66 ± 0.06	41.88 ± 0.08	—	41.70 ± 0.05
115	J235156.13-010913.34	0.174	41.73 ± 0.01	42.23 ± 0.01	42.20 ± 0.00	41.57 ± 0.01

Notes: Columns (4)–(7) list the logarithmic luminosities of the narrow emission-line components. A “—” entry indicates that the corresponding line is not covered or cannot be reliably measured. $\text{H}\alpha$ is generally outside the SDSS spectral coverage for sources at $z \gtrsim 0.4$.

Table A2. Emission-line measurements for PG quasars with SDSS Spectra.

No.	Name	z	$\log L_{[\text{O II}]}$ (erg s^{-1})	$\log L_{[\text{O III}]}$ (erg s^{-1})	$\log L_{\text{H}\beta}$ (erg s^{-1})	$\log L_{\text{H}\alpha}$ (erg s^{-1})
1	PG0157+001	0.164	41.41 ± 0.02	42.26 ± 0.01	42.55 ± 0.00	41.60 ± 0.02
2	PG0921+525	0.035	41.21 ± 0.00	41.80 ± 0.01	41.69 ± 0.01	40.80 ± 0.01

Table A2. Cont.

No.	Name	z	$\log L_{[\text{O II}]}$ (erg s^{-1})	$\log L_{[\text{O III}]}$ (erg s^{-1})	$\log L_{\text{H}\beta}$ (erg s^{-1})	$\log L_{\text{H}\delta}$ (erg s^{-1})
3	PG0923+201	0.190	41.18 ± 0.11	41.32 ± 0.05	40.99 ± 0.35	41.26 ± 0.11
4	PG0934+013	0.050	40.56 ± 0.02	41.44 ± 0.01	41.78 ± 0.00	40.86 ± 0.02
5	PG0947+396	0.206	41.32 ± 0.07	42.08 ± 0.01	42.36 ± 0.01	41.33 ± 0.04
6	PG1001+054	0.161	41.11 ± 0.05	41.81 ± 0.01	42.81 ± 0.00	42.06 ± 0.01
7	PG1004+130	0.240	41.53 ± 0.12	42.43 ± 0.01	42.50 ± 0.01	41.95 ± 0.03
8	PG1022+519	0.045	40.34 ± 0.03	40.89 ± 0.01	41.56 ± 0.05	40.70 ± 0.02
9	PG1048+342	0.167	41.11 ± 0.04	41.96 ± 0.01	42.55 ± 0.00	41.33 ± 0.02
10	PG1049-005	0.357	42.45 ± 0.02	43.36 ± 0.01	41.98 ± 0.11	42.31 ± 0.02
11	PG1114+445	0.144	41.41 ± 0.04	42.25 ± 0.04	42.44 ± 0.01	41.64 ± 0.05
12	PG1115+407	0.154	41.17 ± 0.06	41.80 ± 0.01	42.84 ± 0.01	42.06 ± 0.02
13	PG1121+422	0.234	41.36 ± 0.05	41.90 ± 0.14	43.11 ± 0.19	42.37 ± 0.07
14	PG1151+117	0.176	41.25 ± 0.06	41.91 ± 0.08	42.02 ± 0.04	41.35 ± 0.17
15	PG1202+281	0.165	41.69 ± 0.01	42.32 ± 0.01	42.22 ± 0.01	41.59 ± 0.01
16	PG1229+204	0.064	40.81 ± 0.02	41.19 ± 0.02	41.81 ± 0.01	41.18 ± 0.01
17	PG1244+026	0.048	40.50 ± 0.03	40.40 ± 0.44	41.72 ± 0.01	41.26 ± 0.04
18	PG1259+593	0.472	–	–	–	–
19	PG1309+355	0.184	41.44 ± 0.07	42.38 ± 0.02	42.63 ± 0.01	41.79 ± 0.04
20	PG1341+258	0.087	40.73 ± 0.04	41.41 ± 0.05	41.69 ± 0.01	40.66 ± 0.05
21	PG1415+451	0.114	40.41 ± 0.08	41.25 ± 0.06	42.25 ± 0.01	–
22	PG1425+267	0.366	42.16 ± 0.02	42.87 ± 0.02	42.67 ± 0.01	42.01 ± 0.02
23	PG1427+480	0.221	41.45 ± 0.03	42.44 ± 0.01	42.95 ± 0.00	41.92 ± 0.02
24	PG1444+407	0.267	40.98 ± 0.33	42.17 ± 0.01	43.08 ± 0.01	42.42 ± 0.02
25	PG1512+370	0.371	42.16 ± 0.03	42.90 ± 0.02	42.77 ± 0.01	42.08 ± 0.02
26	PG1534+580	0.030	40.63 ± 0.01	41.60 ± 0.01	41.36 ± 0.01	40.61 ± 0.01
27	PG1543+489	0.400	41.68 ± 0.05	41.89 ± 0.03	43.48 ± 0.00	42.75 ± 0.01
28	PG1552+085	0.119	40.43 ± 0.12	41.56 ± 0.01	42.70 ± 0.01	42.01 ± 0.01
29	PG1612+261	0.131	42.24 ± 0.01	43.00 ± 0.03	42.83 ± 0.00	42.01 ± 0.01
30	PG1704+608	0.371	42.53 ± 0.02	43.36 ± 0.01	45.80 ± 0.22	42.47 ± 0.01
31	PG2233+134	0.325	41.51 ± 0.07	42.58 ± 0.01	43.32 ± 0.00	42.62 ± 0.05

Notes: The columns are the same as in Table A1, but for the PG quasars with available SDSS spectra. Columns (4)–(7) give the logarithmic luminosities of the narrow emission-line components.

Table A3. Emission-line measurements for newly observed PG quasars.

No.	Name	z	$\log L_{[\text{O II}]}$ (erg s^{-1})	$\log L_{[\text{O III}]}$ (erg s^{-1})	$\log L_{\text{H}\beta}$ (erg s^{-1})	$\log L_{\text{H}\delta}$ (erg s^{-1})	Instrument
1	PG0003+158	0.450	42.33 ± 0.07	43.60 ± 0.01	–	43.00 ± 0.02	LJT
2	PG0003+199	0.025	40.82 ± 0.03	41.72 ± 0.00	42.29 ± 0.00	41.53 ± 0.01	HCT
3	PG0007+106	0.089	41.83 ± 0.01	42.57 ± 0.00	42.11 ± 0.01	41.53 ± 0.01	HCT
4	PG0026+129	0.142	41.57 ± 0.02	41.36 ± 0.04	42.05 ± 0.02	–	XLT
5	PG0043+039	0.384	41.83 ± 0.08	41.87 ± 0.14	–	–	XLT
6	PG0049+171	0.064	41.07 ± 0.04	42.11 ± 0.01	42.19 ± 0.00	41.38 ± 0.01	HCT
7	PG0050+124	0.061	40.28 ± 0.20	41.24 ± 0.05	42.61 ± 0.00	41.71 ± 0.01	LJT
8	PG0052+251	0.155	41.98 ± 0.02	42.88 ± 0.01	42.72 ± 0.01	41.96 ± 0.02	LJT
9	PG0804+761	0.100	41.68 ± 0.05	41.95 ± 0.04	43.00 ± 0.01	42.29 ± 0.01	LJT
10	PG0838+770	0.131	40.56 ± 0.20	41.35 ± 0.14	42.08 ± 0.01	41.34 ± 0.06	LJT
11	PG0844+349	0.064	40.87 ± 0.15	41.69 ± 0.01	42.28 ± 0.00	41.47 ± 0.02	LJT
12	PG0923+129	0.029	40.60 ± 0.05	40.97 ± 0.06	41.87 ± 0.01	41.02 ± 0.02	LJT
13	PG0953+414	0.239	–	42.19 ± 0.03	43.32 ± 0.11	–	XLT
14	PG1011-040	0.058	40.09 ± 0.12	41.16 ± 0.01	41.30 ± 0.01	40.84 ± 0.02	HCT
15	PG1012+008	0.185	41.78 ± 0.03	42.16 ± 0.02	42.36 ± 0.02	41.92 ± 0.03	HCT
16	PG1048-090	0.344	42.14 ± 0.16	43.33 ± 0.01	–	42.59 ± 0.03	XLT
17	PG1100+772	0.313	42.52 ± 0.02	43.31 ± 0.02	42.41 ± 0.28	42.39 ± 0.03	XLT

Table A3. Cont.

No.	Name	z	$\log L_{[\text{O II}]}$ (erg s^{-1})	$\log L_{[\text{O III}]}$ (erg s^{-1})	$\log L_{\text{H}\alpha}$ (erg s^{-1})	$\log L_{\text{H}\beta}$ (erg s^{-1})	Instrument
18	PG1103-006	0.425	42.08 ± 0.13	42.65 ± 0.16	—	42.30 ± 0.16	XLT
19	PG1116+215	0.177	41.46 ± 0.23	40.92 ± 0.82	43.15 ± 0.01	42.54 ± 0.02	XLT
20	PG1119+120	0.049	—	—	42.10 ± 0.01	41.08 ± 0.05	LJT
21	PG1126-041	0.060	40.83 ± 0.06	41.20 ± 0.06	41.71 ± 0.01	41.59 ± 0.01	HCT
22	PG1149-110	0.049	40.77 ± 0.02	39.64 ± 5.52	39.40 ± 0.52	39.59 ± 0.30	HCT
23	PG1211+143	0.085	41.35 ± 0.03	—	43.01 ± 0.00	—	LJT
24	PG1216+069	0.334	42.21 ± 0.14	42.96 ± 0.12	43.11 ± 0.15	42.38 ± 0.12	XLT
25	PG1226+023	0.158	42.40 ± 0.09	41.38 ± 0.45	43.61 ± 0.01	43.11 ± 0.02	XLT
26	PG1307+085	0.155	41.88 ± 0.03	42.75 ± 0.02	42.57 ± 0.01	42.00 ± 0.03	XLT
27	PG1310-108	0.035	40.47 ± 0.05	41.61 ± 0.01	41.43 ± 0.01	40.66 ± 0.02	LJT
28	PG1322+659	0.168	41.44 ± 0.07	42.10 ± 0.01	42.49 ± 0.01	40.84 ± 0.58	XLT
29	PG1351+640	0.087	41.69 ± 0.04	42.59 ± 0.01	42.72 ± 0.00	42.10 ± 0.01	HCT
30	PG1352+183	0.158	40.84 ± 0.14	40.94 ± 0.09	—	—	XLT
31	PG1354+213	0.300	41.88 ± 0.07	—	42.62 ± 0.02	42.08 ± 0.07	XLT
32	PG1402+261	0.164	41.65 ± 0.07	42.06 ± 0.02	42.97 ± 0.00	42.16 ± 0.03	XLT
33	PG1411+442	0.089	41.18 ± 0.24	42.28 ± 0.01	42.70 ± 0.01	42.07 ± 0.03	HCT
34	PG1416-129	0.129	41.60 ± 0.01	41.93 ± 0.06	41.95 ± 0.01	41.34 ± 0.03	LJT
35	PG1426+015	0.086	41.14 ± 0.18	41.37 ± 0.21	41.82 ± 0.04	40.65 ± 0.45	LJT
36	PG1440+356	0.077	41.08 ± 0.10	41.26 ± 0.27	42.61 ± 0.01	41.67 ± 0.03	LJT
37	PG1448+273	0.065	41.35 ± 0.04	41.66 ± 0.19	41.65 ± 0.05	41.55 ± 0.04	LJT
38	PG1501+106	0.036	41.09 ± 0.02	42.00 ± 0.00	41.82 ± 0.00	41.22 ± 0.01	LJT
39	PG1519+226	0.137	40.66 ± 0.13	41.40 ± 0.02	42.58 ± 0.00	41.65 ± 0.04	LJT
40	PG1535+547	0.038	—	40.86 ± 0.01	42.00 ± 0.00	41.23 ± 0.01	LJT
41	PG1545+210	0.266	42.23 ± 0.04	43.20 ± 0.02	43.15 ± 0.01	42.52 ± 0.03	LJT
42	PG1613+658	0.129	41.71 ± 0.03	42.24 ± 0.02	41.83 ± 0.03	40.57 ± 0.37	LJT
43	PG1617+175	0.114	41.00 ± 0.07	—	42.24 ± 0.01	41.09 ± 0.09	DBSP
44	PG1626+554	0.133	40.63 ± 0.15	—	42.37 ± 0.01	41.58 ± 0.07	LJT
45	PG1700+518	0.282	—	—	43.65 ± 0.00	—	LJT
46	PG2112+059	0.466	42.23 ± 0.05	42.63 ± 0.02	—	41.41 ± 0.39	XLT
47	PG2130+099	0.061	41.02 ± 0.03	41.91 ± 0.01	42.75 ± 0.00	41.94 ± 0.01	LJT
48	PG2209+184	0.070	40.98 ± 0.09	40.57 ± 0.63	—	—	HCT
49	PG2214+139	0.067	41.15 ± 0.10	41.28 ± 0.02	41.89 ± 0.01	—	HCT
50	PG2251+113	0.323	42.42 ± 0.03	43.32 ± 0.01	43.21 ± 0.01	42.65 ± 0.02	XLT
51	PG2304+042	0.042	40.64 ± 0.05	41.59 ± 0.01	41.29 ± 0.03	40.51 ± 0.04	HCT
52	PG2308+098	0.432	42.00 ± 0.10	43.08 ± 0.04	—	42.57 ± 0.03	XLT

Notes: Columns (4)–(7) list the logarithmic luminosities of the narrow emission-line components. The last column gives the instrument used for the optical spectroscopic observation.

Table A4. Extinction corrections and [O II] star formation rates.

Sample ID	Name	z	$\log \text{SFR}_{\text{FIR}}$ ($M_{\odot} \text{ yr}^{-1}$)	$E(B - V)_{\text{gas}}$	$\log L_{[\text{O II}]}^{\text{corr}}$ (erg s^{-1})	$\log L_{[\text{O III}]}^{\text{corr}}$ (erg s^{-1})	$\log \text{SFR}_{[\text{O II}]}$ ($M_{\odot} \text{ yr}^{-1}$)	$\log \text{SFR}_{[\text{O II}]}^{\text{noAGNcorr}}$ ($M_{\odot} \text{ yr}^{-1}$)
1	J001115.69+011459.23	0.580	2.34 ± 0.15	0.86 ± 0.04	43.20 ± 0.17	42.47 ± 0.42	2.10 ± 0.18	2.11 ± 0.17
2	J002421.84+002508.49	0.599	2.12 ± 0.15	0.59 ± 0.05	42.50 ± 0.20	42.32 ± 0.15	1.30 ± 0.21	1.34 ± 0.20
3	J005905.51+000651.67	0.719	2.00 ± 0.38	0.72 ± 0.11	43.80 ± 0.23	44.10 ± 0.17	2.58 ± 0.24	2.69 ± 0.23
4	J011430.24+000420.94	0.455	1.43 ± 0.14	0.89 ± 0.02	42.72 ± 0.12	41.50 ± 1.42	1.60 ± 0.13	1.61 ± 0.12
5	J011536.92-000011.01	0.532	1.72 ± 0.14	0.90 ± 0.00	43.35 ± 0.04	43.46 ± 0.02	2.17 ± 0.04	2.24 ± 0.04
6	J013354.30+011034.00	0.704	1.92 ± 0.24	0.67 ± 0.12	43.04 ± 0.28	42.79 ± 0.28	1.90 ± 0.29	1.93 ± 0.28
7	J014003.99-002541.77	0.770	2.40 ± 0.19	0.53 ± 0.09	43.07 ± 0.20	42.62 ± 0.39	1.95 ± 0.21	1.97 ± 0.20
8	J015243.15+002039.65	0.578	1.48 ± 0.26	0.89 ± 0.02	43.45 ± 0.10	43.38 ± 0.09	2.29 ± 0.12	2.33 ± 0.10
9	J015950.25+002340.82	0.163	2.40 ± 0.12	0.91 ± 0.04	43.04 ± 0.09	43.51 ± 0.06	1.74 ± 0.11	1.92 ± 0.09
10	J021859.87+002855.80	0.351	1.96 ± 0.18	0.37 ± 0.03	42.06 ± 0.11	42.07 ± 0.08	0.83 ± 0.12	0.88 ± 0.11
11	J022430.61-000038.91	0.431	1.41 ± 0.13	0.64 ± 0.11	42.63 ± 0.23	42.31 ± 0.35	1.34 ± 0.24	1.37 ± 0.23
12	J090026.51+204158.70	0.706	2.35 ± 0.06	0.20 ± 0.01	42.34 ± 0.10	42.45 ± 1.67	1.16 ± 0.39	1.23 ± 0.10
13	J090158.88+002313.87	0.196	1.52 ± 0.09	0.42 ± 0.06	42.01 ± 0.13	42.32 ± 0.13	0.65 ± 0.15	0.76 ± 0.13

Table A4. Cont.

Sample ID	Name	z	$\log \text{SFR}_{\text{FIR}}$ ($M_{\odot} \text{ yr}^{-1}$)	$E(B - V)_{\text{gas}}$	$\log L_{[\text{OII}]}$ (erg s^{-1})	$\log L_{[\text{OIII}]}$ (erg s^{-1})	$\log \text{SFR}_{[\text{OII}]}$ ($M_{\odot} \text{ yr}^{-1}$)	$\log \text{SFR}_{[\text{OII}]}^{\text{noAGNcorr}}$ ($M_{\odot} \text{ yr}^{-1}$)
14	J090933.49+425346.51	0.670	2.46 ± 0.14	0.75 ± 0.02	44.05 ± 0.05	43.28 ± 0.16	2.93 ± 0.05	2.94 ± 0.05
15	J092159.40+450912.38	0.235	1.88 ± 0.07	0.59 ± 0.02	42.98 ± 0.05	42.02 ± 0.23	1.73 ± 0.05	1.73 ± 0.05
17	J092635.12+072446.43	0.189	1.25 ± 0.11	0.60 ± 0.09	42.55 ± 0.18	42.86 ± 0.13	1.33 ± 0.18	1.44 ± 0.18
18	J094652.58+131953.83	0.133	1.20 ± 0.19	0.64 ± 0.06	43.31 ± 0.13	43.32 ± 0.13	2.11 ± 0.13	2.16 ± 0.13
19	J094745.14+072520.58	0.086	0.04 ± 0.12	0.30 ± 0.05	41.51 ± 0.10	41.97 ± 0.11	0.25 ± 0.13	0.42 ± 0.10
20	J095017.06+215022.41	0.455	1.67 ± 0.07	0.68 ± 0.07	42.19 ± 0.26	42.85 ± 0.11	0.72 ± 0.43	1.03 ± 0.26
21	J095240.17+515249.91	0.554	1.63 ± 0.19	0.32 ± 0.17	41.84 ± 0.45	42.69 ± 0.26	0.11 ± 0.98	0.75 ± 0.45
23	J100043.14+020637.25	0.360	1.14 ± 0.11	0.43 ± 0.05	42.01 ± 0.13	41.98 ± 0.22	0.86 ± 0.15	0.91 ± 0.13
24	J100232.13+023537.33	0.658	1.75 ± 0.11	0.45 ± 0.18	42.45 ± 0.40	42.85 ± 0.30	1.21 ± 0.44	1.35 ± 0.40
25	J100420.13+051300.48	0.160	0.93 ± 0.22	0.52 ± 0.03	42.07 ± 0.10	42.53 ± 0.05	0.76 ± 0.13	0.92 ± 0.10
26	J103651.94+575950.96	0.500	1.65 ± 0.13	0.50 ± 0.11	42.47 ± 0.26	42.34 ± 0.39	1.31 ± 0.29	1.35 ± 0.26
28	J104505.39+561118.34	0.428	1.88 ± 0.10	0.75 ± 0.02	43.76 ± 0.05	43.46 ± 0.07	2.62 ± 0.05	2.65 ± 0.05
29	J104739.49+563507.19	0.303	1.71 ± 0.15	0.59 ± 0.09	42.08 ± 0.25	41.99 ± 0.30	0.88 ± 0.27	0.92 ± 0.25
30	J105106.12+591625.24	0.767	2.03 ± 0.16	0.20 ± 0.10	42.46 ± 0.27	42.73 ± 0.28	1.26 ± 0.34	1.36 ± 0.27
31	J105143.89+335926.71	0.167	0.86 ± 0.11	1.45 ± 0.07	44.03 ± 0.17	44.07 ± 0.20	2.86 ± 0.19	2.92 ± 0.17
32	J105151.44+005117.66	0.359	2.34 ± 0.12	0.44 ± 0.05	43.28 ± 0.12	43.98 ± 0.09	1.84 ± 0.16	2.19 ± 0.12
33	J105705.41+580437.46	0.140	1.17 ± 0.12	0.80 ± 0.05	42.12 ± 0.14	42.46 ± 0.09	0.89 ± 0.16	1.01 ± 0.14
34	J105959.93+574848.17	0.453	1.58 ± 0.18	0.44 ± 0.09	42.19 ± 0.23	42.99 ± 0.16	0.45 ± 0.46	0.96 ± 0.23
36	J111706.40+441333.30	0.144	0.81 ± 0.13	0.60 ± 0.09	42.53 ± 0.20	43.09 ± 0.15	1.22 ± 0.25	1.44 ± 0.20
37	J111830.29+402554.02	0.154	1.41 ± 0.05	0.36 ± 0.04	41.80 ± 0.12	42.14 ± 0.14	0.48 ± 0.17	0.60 ± 0.12
38	J112019.62+130320.10	0.314	1.46 ± 0.12	0.52 ± 0.08	42.24 ± 0.18	42.26 ± 0.34	1.03 ± 0.22	1.08 ± 0.18
39	J112048.99+133821.91	0.513	1.60 ± 0.05	0.34 ± 0.12	42.15 ± 0.28	42.30 ± 0.24	0.85 ± 0.30	0.93 ± 0.28
40	J112759.26+360207.00	0.667	2.19 ± 0.26	0.63 ± 0.14	43.03 ± 0.29	43.43 ± 0.22	1.79 ± 0.31	1.92 ± 0.29
41	J114121.76+014803.58	0.382	1.82 ± 0.14	0.48 ± 0.07	42.50 ± 0.17	42.75 ± 0.13	1.25 ± 0.19	1.35 ± 0.17
42	J114933.88+222227.07	0.554	1.66 ± 0.07	0.21 ± 0.03	41.71 ± 0.15	41.56 ± 0.39	0.52 ± 0.19	0.55 ± 0.15
43	J115253.68+000131.90	0.823	2.23 ± 0.25	0.69 ± 0.12	43.20 ± 0.30	43.31 ± 0.25	2.03 ± 0.32	2.09 ± 0.30
44	J120226.76+012915.28	0.150	2.24 ± 0.10	0.85 ± 0.07	42.84 ± 0.15	42.55 ± 0.11	1.65 ± 0.15	1.67 ± 0.15
45	J120312.14+015321.30	0.296	1.45 ± 0.21	0.81 ± 0.08	42.52 ± 0.19	41.92 ± 0.36	1.41 ± 0.20	1.43 ± 0.19
46	J120442.11+275411.80	0.165	1.39 ± 0.14	0.27 ± 0.03	42.19 ± 0.06	42.70 ± 0.05	0.80 ± 0.07	0.99 ± 0.06
47	J120734.63+150643.69	0.750	2.07 ± 0.24	0.35 ± 0.09	42.41 ± 0.25	43.05 ± 0.31	1.02 ± 0.49	1.30 ± 0.25
48	J121037.81+053805.88	0.436	1.63 ± 0.09	0.43 ± 0.08	41.90 ± 0.25	42.36 ± 0.18	0.53 ± 0.34	0.69 ± 0.25
49	J121728.37+065110.00	0.589	1.94 ± 0.42	0.13 ± 0.03	41.79 ± 0.15	42.36 ± 0.13	0.47 ± 0.27	0.69 ± 0.15
50	J121836.71+155908.43	0.766	2.15 ± 0.11	0.37 ± 0.05	42.70 ± 0.15	42.85 ± 0.17	1.48 ± 0.18	1.55 ± 0.15
51	J121906.57+160243.06	0.761	2.28 ± 0.21	0.63 ± 0.10	43.28 ± 0.22	42.41 ± 0.71	2.16 ± 0.23	2.17 ± 0.22
52	J121945.03+082117.95	0.228	1.78 ± 0.13	0.96 ± 0.07	42.98 ± 0.15	42.20 ± 0.19	1.84 ± 0.15	1.85 ± 0.15
53	J121946.54+145259.37	0.401	1.17 ± 0.31	0.05 ± 0.01	41.47 ± 0.06	41.97 ± 0.05	0.00 ± 0.11	0.19 ± 0.06
54	J122011.88+020342.21	0.240	1.43 ± 0.32	0.71 ± 0.04	43.28 ± 0.09	43.78 ± 0.06	1.99 ± 0.11	2.17 ± 0.09
55	J122026.72+062748.20	0.349	1.64 ± 0.09	0.83 ± 0.05	42.22 ± 0.22	42.43 ± 0.21	1.05 ± 0.27	1.13 ± 0.22
56	J122102.50+155447.04	0.229	1.18 ± 0.21	0.57 ± 0.12	42.23 ± 0.25	42.60 ± 0.17	0.90 ± 0.27	1.03 ± 0.25
57	J122102.95+000733.74	0.366	2.21 ± 0.10	0.46 ± 0.05	42.81 ± 0.10	42.28 ± 0.09	1.70 ± 0.10	1.71 ± 0.10
58	J122106.50+114625.46	0.340	1.52 ± 0.07	0.59 ± 0.08	42.50 ± 0.18	42.62 ± 0.16	1.32 ± 0.19	1.39 ± 0.18
59	J122307.37+002124.73	0.804	2.23 ± 0.23	0.88 ± 0.03	43.74 ± 0.09	43.91 ± 0.06	2.55 ± 0.10	2.63 ± 0.09
60	J122312.17+095017.72	0.277	1.55 ± 0.28	0.39 ± 0.10	41.69 ± 0.27	42.34 ± 0.24	0.22 ± 0.45	0.52 ± 0.27
61	J122317.80+092306.94	0.682	2.76 ± 0.10	0.25 ± 0.02	42.40 ± 0.13	42.20 ± 0.18	1.27 ± 0.15	1.30 ± 0.13
62	J122404.62+045637.94	0.358	1.76 ± 0.24	0.45 ± 0.14	42.64 ± 0.28	42.64 ± 0.30	1.42 ± 0.30	1.47 ± 0.28
63	J122520.13+084450.76	0.535	2.06 ± 0.04	0.68 ± 0.14	43.20 ± 0.32	43.25 ± 0.25	2.03 ± 0.34	2.09 ± 0.32
64	J122526.21+141332.24	0.760	2.35 ± 0.12	0.31 ± 0.10	42.34 ± 0.28	42.37 ± 0.26	1.18 ± 0.30	1.24 ± 0.28
65	J122641.50+055906.81	0.290	1.31 ± 0.18	0.90 ± 0.01	42.78 ± 0.08	42.98 ± 0.17	1.59 ± 0.13	1.67 ± 0.08
66	J122822.10+114606.83	0.365	1.86 ± 0.07	0.39 ± 0.06	42.43 ± 0.13	42.52 ± 0.09	1.26 ± 0.13	1.32 ± 0.13
67	J122839.20+035749.29	0.608	2.20 ± 0.10	0.27 ± 0.02	41.85 ± 0.12	41.94 ± 0.39	0.68 ± 0.19	0.75 ± 0.12
68	J123436.54+123918.66	0.777	2.44 ± 0.16	0.44 ± 0.06	42.93 ± 0.13	43.17 ± 0.15	1.73 ± 0.15	1.82 ± 0.13
69	J123800.92+621336.09	0.440	1.84 ± 0.16	0.47 ± 0.05	42.03 ± 0.18	42.27 ± 0.10	0.80 ± 0.20	0.89 ± 0.18
71	J125257.36+331555.81	0.367	1.32 ± 0.16	0.89 ± 0.02	43.15 ± 0.07	42.86 ± 0.09	2.02 ± 0.08	2.04 ± 0.07
72	J125317.57+310550.64	0.782	2.39 ± 0.21	0.08 ± 0.08	42.45 ± 0.19	43.02 ± 0.19	1.11 ± 0.28	1.34 ± 0.19
73	J125553.05+272405.23	0.316	1.03 ± 0.14	0.86 ± 0.04	42.64 ± 0.14	42.83 ± 0.09	1.44 ± 0.16	1.53 ± 0.14
75	J125711.97+274216.45	0.793	2.09 ± 0.14	0.17 ± 0.03	42.08 ± 0.10	42.15 ± 0.40	0.91 ± 0.16	0.97 ± 0.10
76	J125757.23+322929.29	0.806	2.60 ± 0.26	0.74 ± 0.05	42.76 ± 0.32	43.46 ± 0.10	1.31 ± 0.61	1.65 ± 0.32
77	J130622.96+225752.95	0.758	2.06 ± 0.31	0.47 ± 0.09	42.57 ± 0.24	41.59 ± 1.27	1.47 ± 0.25	1.48 ± 0.24
78	J130947.00+081948.23	0.154	1.00 ± 0.22	0.64 ± 0.04	42.91 ± 0.11	43.47 ± 0.09	1.61 ± 0.14	1.82 ± 0.11
79	J131247.97+250756.71	0.425	2.21 ± 0.21	0.78 ± 0.09	43.56 ± 0.19	43.72 ± 0.14	2.38 ± 0.19	2.45 ± 0.19
80	J131312.13+284730.01	0.259	0.93 ± 0.17	0.89 ± 0.03	42.48 ± 0.11	42.08 ± 0.16	1.35 ± 0.12	1.37 ± 0.11
81	J131531.66+265414.73	0.620	1.92 ± 0.21	0.61 ± 0.12	42.72 ± 0.30	43.12 ± 0.27	1.42 ± 0.37	1.56 ± 0.30
82	J132919.84+250626.45	0.781	2.22 ± 0.20	0.40 ± 0.10	42.13 ± 0.31	42.13 ± 0.39	0.96 ± 0.35	1.01 ± 0.31
84	J133442.29+320939.60	0.649	2.39 ± 0.30	0.20 ± 0.01	41.61 ± 0.23	41.79 ± 0.23	0.44 ± 0.31	0.52 ± 0.23
85	J134356.74+253847.69	0.086	0.45 ± 0.08	0.91 ± 0.07	42.33 ± 0.18	42.75 ± 0.13	1.02 ± 0.21	1.17 ± 0.18
86	J135632.80+210352.35	0.301	1.28 ± 0.21	0.29 ± 0.09	42.12 ± 0.23	42.97 ± 0.13	0.19 ± 0.46	0.82 ± 0.23
88	J140655.66+015712.88	0.427	2.10 ± 0.19	0.36 ± 0.08	42.08 ± 0.20	42.43 ± 0.14	0.81 ± 0.22	0.93 ± 0.20
90	J141637.45+003352.28	0.434	2.32 ± 0.19	0.69 ± 0.04	43.11 ± 0.10	43.51 ± 0.10	1.86 ± 0.12	1.99 ± 0.10

Table A4. Cont.

Sample ID	Name	z	$\log \text{SFR}_{\text{FIR}}$ ($M_{\odot} \text{ yr}^{-1}$)	$E(B - V)_{\text{gas}}$	$\log L_{[\text{O II}]}$ (erg s^{-1})	$\log L_{[\text{O III}]}$ (erg s^{-1})	$\log \text{SFR}_{[\text{O II}]}$ ($M_{\odot} \text{ yr}^{-1}$)	$\log \text{SFR}_{[\text{O II}]}^{\text{noAGNcorr}}$ ($M_{\odot} \text{ yr}^{-1}$)
91	J141644.62+190541.92	0.365	1.67 ± 0.11	0.37 ± 0.06	41.96 ± 0.15	40.67 ± 0.15	0.70 ± 0.15	0.70 ± 0.15
92	J141700.83+445606.39	0.113	0.73 ± 0.27	0.68 ± 0.07	41.72 ± 0.21	42.20 ± 0.14	0.43 ± 0.26	0.61 ± 0.21
93	J142052.44+525622.42	0.677	2.17 ± 0.29	0.21 ± 0.10	42.07 ± 0.26	42.79 ± 0.15	0.60 ± 0.40	0.97 ± 0.26
94	J142648.78+005323.24	0.220	1.40 ± 0.11	0.64 ± 0.08	42.69 ± 0.16	42.80 ± 0.27	1.51 ± 0.19	1.58 ± 0.16
95	J142710.94+013023.22	0.704	2.45 ± 0.27	0.47 ± 0.09	43.19 ± 0.21	42.52 ± 0.39	2.08 ± 0.22	2.09 ± 0.21
96	J142753.79+345248.35	0.514	1.58 ± 0.11	0.27 ± 0.05	41.76 ± 0.21	42.12 ± 0.11	0.41 ± 0.27	0.54 ± 0.21
97	J142918.15+592106.65	0.739	2.70 ± 0.14	0.73 ± 0.03	43.59 ± 0.08	43.87 ± 0.12	2.38 ± 0.10	2.48 ± 0.08
98	J142943.07+474726.23	0.221	1.27 ± 0.30	0.94 ± 0.03	43.28 ± 0.08	43.81 ± 0.05	1.96 ± 0.10	2.16 ± 0.08
99	J143624.81-002905.36	0.325	1.66 ± 0.31	0.95 ± 0.05	42.70 ± 0.20	43.30 ± 0.09	1.23 ± 0.29	1.47 ± 0.20
100	J144231.82+014353.43	0.280	0.99 ± 0.14	0.84 ± 0.08	42.36 ± 0.23	42.15 ± 0.23	1.22 ± 0.24	1.25 ± 0.23
102	J145108.76+270926.92	0.064	0.63 ± 0.21	0.31 ± 0.03	41.61 ± 0.07	42.21 ± 0.05	0.26 ± 0.09	0.51 ± 0.07
103	J145538.73+002238.06	0.434	1.71 ± 0.21	0.90 ± 0.01	42.51 ± 0.18	42.86 ± 0.07	1.27 ± 0.25	1.40 ± 0.18
104	J152114.26+222743.87	0.136	1.02 ± 0.16	0.88 ± 0.04	42.51 ± 0.12	42.83 ± 0.06	1.28 ± 0.13	1.39 ± 0.12
105	J154530.24+484608.98	0.399	2.51 ± 0.08	0.48 ± 0.02	42.56 ± 0.09	42.55 ± 0.06	1.42 ± 0.10	1.47 ± 0.09
107	J161413.20+260416.21	0.131	0.91 ± 0.26	0.65 ± 0.02	43.45 ± 0.05	43.90 ± 0.06	2.08 ± 0.06	2.24 ± 0.05
108	J163352.34+402115.66	0.782	2.14 ± 0.08	0.50 ± 0.11	42.72 ± 0.26	42.81 ± 0.33	1.55 ± 0.29	1.61 ± 0.26
109	J163915.81+412833.70	0.690	1.84 ± 0.05	0.45 ± 0.10	42.51 ± 0.32	42.89 ± 0.18	1.26 ± 0.38	1.40 ± 0.32
110	J171033.22+584456.86	0.281	1.53 ± 0.15	0.80 ± 0.07	42.49 ± 0.18	42.36 ± 0.14	1.36 ± 0.19	1.39 ± 0.18
111	J171352.43+584201.25	0.521	2.11 ± 0.20	0.29 ± 0.08	42.58 ± 0.17	43.15 ± 0.13	1.26 ± 0.21	1.48 ± 0.17
112	J220759.39+001722.62	0.368	1.40 ± 0.21	0.21 ± 0.03	41.22 ± 0.13	40.99 ± 0.24	0.03 ± 0.15	0.06 ± 0.13
113	J223607.68+134355.32	0.326	1.64 ± 0.13	0.41 ± 0.13	42.26 ± 0.30	43.15 ± 0.20	0.27 ± 0.61	1.07 ± 0.30
114	J233741.32+001743.77	0.762	2.08 ± 0.05	0.43 ± 0.02	42.47 ± 0.10	42.47 ± 0.11	1.25 ± 0.12	1.30 ± 0.10
115	J235156.13-010913.34	0.174	1.46 ± 0.15	0.26 ± 0.02	42.50 ± 0.05	42.59 ± 0.04	0.96 ± 0.06	1.09 ± 0.05
117	PG0003+199	0.025	-0.09 ± 0.24	0.53 ± 0.01	41.81 ± 0.05	42.45 ± 0.02	0.36 ± 0.08	0.64 ± 0.05
118	PG0007+106	0.089	0.94 ± 0.17	0.18 ± 0.03	42.16 ± 0.07	42.82 ± 0.04	0.75 ± 0.09	1.05 ± 0.07
119	PG0026+129	0.142	0.49 ± 0.15	0.24 ± 0.08	42.03 ± 0.17	41.69 ± 0.15	0.89 ± 0.18	0.92 ± 0.17
120	PG0043+039	0.384	1.24 ± 0.04	0.33 ± 0.13	42.44 ± 0.32	42.33 ± 0.32	1.31 ± 0.34	1.35 ± 0.32
121	PG0049+171	0.064	-0.37 ± 0.29	0.63 ± 0.02	42.25 ± 0.08	42.99 ± 0.03	0.56 ± 0.14	0.95 ± 0.08
123	PG0052+251	0.155	1.13 ± 0.17	0.53 ± 0.05	42.96 ± 0.11	43.61 ± 0.08	1.57 ± 0.14	1.85 ± 0.11
124	PG0157+001	0.164	2.53 ± 0.10	0.90 ± 0.04	43.09 ± 0.09	43.51 ± 0.06	1.85 ± 0.10	1.99 ± 0.09
125	PG0804+761	0.100	0.73 ± 0.27	0.45 ± 0.04	42.52 ± 0.12	42.57 ± 0.09	1.37 ± 0.13	1.42 ± 0.12
128	PG0921+525	0.035	-0.32 ± 0.15	0.79 ± 0.03	42.67 ± 0.06	42.89 ± 0.06	1.42 ± 0.06	1.51 ± 0.06
129	PG0923+201	0.190	0.84 ± 0.16	0.61 ± 0.13	42.32 ± 0.35	42.17 ± 0.22	1.18 ± 0.36	1.21 ± 0.35
130	PG0923+129	0.029	0.56 ± 0.16	0.70 ± 0.04	41.90 ± 0.12	41.94 ± 0.11	0.75 ± 0.14	0.81 ± 0.12
131	PG0934+013	0.050	0.47 ± 0.16	0.84 ± 0.04	42.12 ± 0.08	42.60 ± 0.06	0.76 ± 0.10	0.93 ± 0.08
132	PG0947+396	0.206	1.23 ± 0.24	1.06 ± 0.09	43.30 ± 0.23	43.56 ± 0.14	2.09 ± 0.25	2.19 ± 0.23
134	PG1001+054	0.161	0.84 ± 0.26	0.52 ± 0.03	42.08 ± 0.10	42.54 ± 0.05	0.78 ± 0.13	0.94 ± 0.10
135	PG1004+130	0.240	1.68 ± 0.18	0.13 ± 0.07	41.77 ± 0.25	42.61 ± 0.11	0.06 ± 0.68	0.67 ± 0.25
137	PG1012+008	0.185	1.36 ± 0.17	0.62 ± 0.16	42.95 ± 0.32	43.03 ± 0.24	1.77 ± 0.33	1.84 ± 0.32
138	PG1022+519	0.045	0.33 ± 0.04	0.72 ± 0.11	41.70 ± 0.24	41.89 ± 0.16	0.51 ± 0.25	0.59 ± 0.24
139	PG1048+342	0.167	0.90 ± 0.17	1.45 ± 0.05	43.80 ± 0.13	43.97 ± 0.07	2.62 ± 0.14	2.69 ± 0.13
141	PG1049-005	0.357	2.31 ± 0.15	0.45 ± 0.14	43.28 ± 0.27	43.99 ± 0.20	1.84 ± 0.30	2.19 ± 0.27
142	PG1100+772	0.313	1.65 ± 0.20	0.43 ± 0.13	43.32 ± 0.26	43.90 ± 0.20	1.99 ± 0.29	2.22 ± 0.26
143	PG1103-006	0.425	1.36 ± 0.25	0.54 ± 0.13	43.09 ± 0.36	43.40 ± 0.34	1.81 ± 0.45	1.92 ± 0.36
144	PG1114+445	0.144	0.79 ± 0.14	0.61 ± 0.09	42.55 ± 0.21	43.10 ± 0.16	1.23 ± 0.26	1.44 ± 0.21
145	PG1115+407	0.154	1.42 ± 0.04	0.58 ± 0.03	42.24 ± 0.13	42.59 ± 0.06	0.97 ± 0.15	1.10 ± 0.13
146	PG1116+215	0.177	1.16 ± 0.02	0.45 ± 0.14	42.30 ± 0.49	41.54 ± 1.01	1.15 ± 0.51	1.16 ± 0.49
148	PG1121+422	0.234	0.72 ± 0.26	0.13 ± 0.08	41.60 ± 0.21	42.07 ± 0.26	0.33 ± 0.31	0.50 ± 0.21
149	PG1126-041	0.060	0.96 ± 0.22	0.42 ± 0.13	41.61 ± 0.30	41.78 ± 0.24	0.44 ± 0.33	0.51 ± 0.30
150	PG1149-110	0.049	0.43 ± 0.30	0.22 ± 0.10	41.19 ± 0.20	39.95 ± 5.65	0.07 ± 0.24	0.08 ± 0.20
151	PG1151+117	0.176	0.47 ± 0.03	0.54 ± 0.13	42.26 ± 0.31	42.66 ± 0.26	1.03 ± 0.36	1.17 ± 0.31
152	PG1202+281	0.165	1.33 ± 0.16	0.28 ± 0.03	42.21 ± 0.06	42.71 ± 0.05	0.89 ± 0.07	1.07 ± 0.06
154	PG1216+069	0.334	0.95 ± 0.28	0.21 ± 0.11	42.61 ± 0.35	43.25 ± 0.28	1.22 ± 0.60	1.51 ± 0.35
155	PG1226+023	0.158	1.74 ± 0.02	0.05 ± 0.00	42.50 ± 0.09	41.45 ± 0.45	1.38 ± 0.10	1.39 ± 0.09
156	PG1229+204	0.064	0.67 ± 0.25	0.28 ± 0.03	41.33 ± 0.07	41.57 ± 0.06	0.13 ± 0.08	0.22 ± 0.07
160	PG1307+085	0.155	0.88 ± 0.29	0.16 ± 0.06	42.18 ± 0.14	42.97 ± 0.09	0.57 ± 0.24	1.07 ± 0.14
161	PG1309+355	0.184	1.22 ± 0.24	0.69 ± 0.07	42.72 ± 0.20	43.33 ± 0.12	1.36 ± 0.27	1.61 ± 0.20
163	PG1322+659	0.168	1.15 ± 0.10	0.62 ± 0.13	42.60 ± 0.32	42.96 ± 0.19	1.37 ± 0.34	1.50 ± 0.32
164	PG1341+258	0.087	0.48 ± 0.16	1.07 ± 0.10	42.72 ± 0.23	42.89 ± 0.19	1.55 ± 0.24	1.62 ± 0.23
166	PG1351+640	0.087	1.51 ± 0.25	0.26 ± 0.03	42.17 ± 0.09	42.95 ± 0.05	0.48 ± 0.19	0.94 ± 0.09
167	PG1352+183	0.158	0.37 ± 0.15	0.59 ± 0.13	41.94 ± 0.38	41.75 ± 0.27	0.82 ± 0.40	0.85 ± 0.38
169	PG1402+261	0.164	1.55 ± 0.09	0.63 ± 0.06	42.82 ± 0.18	42.93 ± 0.10	1.61 ± 0.19	1.67 ± 0.18
171	PG1411+442	0.089	0.57 ± 0.14	0.48 ± 0.13	42.08 ± 0.50	42.95 ± 0.20	0.19 ± 1.62	0.92 ± 0.50
172	PG1415+451	0.114	0.92 ± 0.33	0.52 ± 0.09	41.37 ± 0.25	41.97 ± 0.19	0.02 ± 0.36	0.26 ± 0.25
173	PG1416-129	0.129	0.70 ± 0.13	0.23 ± 0.07	42.04 ± 0.14	42.26 ± 0.15	0.80 ± 0.16	0.89 ± 0.14
174	PG1425+267	0.366	1.81 ± 0.09	0.34 ± 0.05	42.80 ± 0.12	43.34 ± 0.09	1.49 ± 0.14	1.70 ± 0.12
175	PG1426+015	0.086	1.17 ± 0.17	0.77 ± 0.09	42.58 ± 0.34	42.44 ± 0.34	1.43 ± 0.38	1.47 ± 0.34
176	PG1427+480	0.221	1.13 ± 0.34	1.06 ± 0.04	43.44 ± 0.10	43.92 ± 0.06	2.13 ± 0.12	2.30 ± 0.10

Table A4. Cont.

Sample ID	Name	z	$\log \text{SFR}_{\text{FIR}}$ ($M_{\odot} \text{ yr}^{-1}$)	$E(B - V)_{\text{gas}}$	$\log L_{[\text{O II}]}$ (erg s^{-1})	$\log L_{[\text{O III}]}$ (erg s^{-1})	$\log \text{SFR}_{[\text{O II}]}$ ($M_{\odot} \text{ yr}^{-1}$)	$\log \text{SFR}_{[\text{O II}]}^{\text{noAGNcorr}}$ ($M_{\odot} \text{ yr}^{-1}$)
178	PG1440+356	0.077	1.33 ± 0.08	0.80 ± 0.10	42.57 ± 0.28	42.37 ± 0.41	1.43 ± 0.31	1.46 ± 0.28
180	PG1448+273	0.065	0.51 ± 0.28	0.23 ± 0.10	41.77 ± 0.24	41.97 ± 0.34	0.60 ± 0.29	0.68 ± 0.24
182	PG1512+370	0.371	1.47 ± 0.27	0.38 ± 0.04	42.87 ± 0.12	43.43 ± 0.08	1.54 ± 0.15	1.76 ± 0.12
183	PG1519+226	0.137	1.01 ± 0.22	0.88 ± 0.08	42.30 ± 0.27	42.62 ± 0.13	1.09 ± 0.32	1.20 ± 0.27
186	PG1543+489	0.400	2.35 ± 0.18	0.47 ± 0.01	42.57 ± 0.08	42.55 ± 0.05	1.42 ± 0.09	1.47 ± 0.08
187	PG1545+210	0.266	0.98 ± 0.03	0.28 ± 0.06	42.74 ± 0.14	43.58 ± 0.09	1.01 ± 0.32	1.62 ± 0.14
189	PG1612+261	0.131	0.95 ± 0.31	0.65 ± 0.02	43.45 ± 0.05	43.90 ± 0.06	2.16 ± 0.07	2.32 ± 0.05
190	PG1613+658	0.129	1.76 ± 0.09	0.38 ± 0.14	42.42 ± 0.29	42.76 ± 0.21	1.19 ± 0.31	1.31 ± 0.29
194	PG1704+608	0.371	2.17 ± 0.10	0.75 ± 0.08	43.93 ± 0.16	44.41 ± 0.12	2.65 ± 0.18	2.82 ± 0.16
195	PG2112+059	0.466	2.13 ± 0.28	0.64 ± 0.11	43.43 ± 0.26	43.52 ± 0.18	2.27 ± 0.27	2.33 ± 0.26
196	PG2130+099	0.061	1.01 ± 0.11	0.65 ± 0.01	42.22 ± 0.05	42.81 ± 0.03	0.78 ± 0.08	1.01 ± 0.05
197	PG2209+184	0.070	0.33 ± 0.08	0.31 ± 0.03	41.56 ± 0.14	41.01 ± 0.67	0.44 ± 0.16	0.45 ± 0.14
198	PG2214+139	0.067	0.20 ± 0.11	0.22 ± 0.04	41.56 ± 0.17	41.59 ± 0.08	0.40 ± 0.19	0.45 ± 0.17
199	PG2233+134	0.325	1.55 ± 0.21	0.43 ± 0.11	42.30 ± 0.27	43.17 ± 0.16	0.38 ± 0.59	1.08 ± 0.27
200	PG2251+113	0.323	1.28 ± 0.10	0.14 ± 0.05	42.69 ± 0.12	43.52 ± 0.08	1.02 ± 0.22	1.58 ± 0.12
201	PG2304+042	0.042	-0.45 ± 0.18	0.57 ± 0.09	41.71 ± 0.22	42.38 ± 0.13	0.28 ± 0.28	0.60 ± 0.22
202	PG2308+098	0.432	1.29 ± 0.06	0.38 ± 0.13	42.72 ± 0.33	43.62 ± 0.21	0.74 ± 1.19	1.61 ± 0.33

Notes: Column (1): sample ID, which follows the original numbering of the full sample of 202 quasars. The IDs are not necessarily consecutive because objects without reliable [O II] or [O III] measurements are omitted. Column (2): source name. Column (3): redshift. Column (4): FIR SFR derived from the cold-dust component of the SED integrated over 8–1000 μm . Column (5): gas color excess used for the internal extinction correction. Columns (6) and (7): extinction-corrected luminosities of [O II] $\lambda 3727$ and [O III] $\lambda 5007$, respectively. Column (8): [O II] SFR after subtracting the AGN narrow-line-region contribution estimated as $0.109L_{[\text{O III}]}$. Column (9): [O II] SFR computed without subtracting this [O III] narrow-line-region contribution.

Note

¹ <https://irsa.ipac.caltech.edu/> (accessed on 24 June 2026).

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